Determinate Indeterminacy: Keynesian Search in Stochastic Overlapping Generations Models

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Abstract

How much freedom remains in agents' beliefs about future wealth once an economy is populated by heterogeneous, finitely-lived generations whose asset holdings evolve endogenously? We study this question in a stochastic overlapping generations model with incomplete labor markets and Keynesian search in the tradition of Roger Farmer. Farmer's representative-agent framework leaves the belief function that closes the model entirely arbitrary. We show this is no longer true: rational expectations require the belief rule to reproduce the endogenous wealth distribution, disciplining an indeterminate steady state. We calibrate a 58-period lifecycle economy with dual labor markets and matching efficiency shocks, solve it via a neural network, and recover equilibrium expectations.

1. Introduction

This paper extends Roger Farmer's work on Keynesian search to a multi-period stochastic overlapping generations model. Farmer's key observation is that in many instances, labor markets are incomplete, in the sense that labor is an experience good. Firms may screen workers to try and find the best matches, but whether or not a worker actually fits can only be determined by putting the worker to work. In Farmer's DSGE implementations of this idea, wages can no longer clear labor markets, and expectations of the representative household become equilibrium objects that close the model. In this framework, however, the steady state indeterminacy generated by the labor market incompleteness assumption extends to expectations: any expectations function will generate a rational expectations equilibrium (REE) by construction. In this paper, we show that this is not the case in a general stochastic overlapping generations (SOLG) model, where the endogenously generated distribution of wealth matters in equilibrium.

In this paper, we ask the following question. When labor-market incompleteness makes employment and output depend on beliefs, how much freedom remains in those beliefs once the economy is populated by finitely lived heterogeneous agents whose wealth distribution evolves endogenously? Our answer is that the steady-state indeterminacy emphasized by Farmer survives, but the associated belief functions are not arbitrary in the SOLG setting. Rational expectations require the belief rule to reproduce the law of motion for the endogenous wealth distribution generated by household saving and portfolio choices.

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The paper makes three contributions. First, it provides a three-period Diamond version of the deterministic OLG model showing how labor-market incompleteness creates steady-state indeterminacy without invoking monetary OLG indeterminacy. Second, it shows how the transition from a deterministic three-period environment to a stochastic environment changes the kind of mechanism one might use to close the model. In the deterministic Farmer-style benchmark, labor-market incompleteness leaves the stationary allocation underdetermined unless an additional restriction – a belief rule, an employment rule, or a bargaining rule – is imposed, and any such restriction is sufficient to close the model. In the SOLG environment, this is no longer true of an arbitrary belief rule: rational expectations requires the temporary equilibrium generated by a candidate belief rule to reproduce the endogenous wealth distribution used in agents' forecasts. This consistency condition disciplines which belief rules are admissible, but the model itself is closed, as we show in our calibration of the exogenous supply side of the economy. Third, the paper develops a neural-network solution method for computing the recursive equilibrium of a calibrated 58-period economy with entry-level labor-market search, matching-efficiency shocks, and endogenous portfolio dynamics.

The argument for the proposition that labor markets are inherently incomplete is based on information asymmetries about the overall value of a match. Empirical evidence for labor market incompleteness goes back at least to Jovanovic (1979) and Mincer and Jovanovic (1981). Both of these papers show that once one controls for the obvious correlations between separation rates and job tenure, there is still a negative structural relationship between these variables. Jovanovic (1979) interprets this as evidence that employment matches, at least at entry levels or for unskilled workers, are experience goods, as opposed to pure search goods. Under this interpretation, employers and workers can only learn the value of an employment match by making it and seeing if it works. Under a pure search goods interpretation, the value of potential matches can be determined once a search meeting has occurred, with good matches accepted and bad ones rejected. This is similar to how matching models are used in search-based models of money. However, under this interpretation, wages do serve to induce greater search effort by workers when firms need more labor, and hence the fundamental mechanism of market clearing is preserved in these models. In experience good models, wages serve a different purpose. As Jovanovic notes, in these models "employers can contract with workers on an individual basis. The employer is then able to reward a worker with whom he matches well by paying the worker relatively more." (Ibid., p. 974) This function of wages is independent of any market-clearing role.

There is additional evidence for the labor-as-an-experience-good hypothesis in Kotlikoff (1979). In this paper, Kotlikoff examines data on the New Orleans slave market between 1804 and 1862. One striking observation in this data is that the sale of individual slaves generally came with a money-back guarantee as to the slave's ability to perform different tasks. As Kotlikoff notes, "In the case of slaves who were not fully guaranteed, the exclusions from the full guarantee are often stated." (Ibid., p. 497) Kotlikoff also notes that these data are generally considered reliable because they were required by law to establish each owner's right to the particular slave being purchased.

Based on this kind of empirical evidence, together with a trove of evidence in the finance literature on the causal role that financial markets play in predicting business cycles, Farmer (2012) introduced an alternative search mechanism which explicitly broke the link between wages and market clearing, which he dubbed (Old) Keynesian search theory. In this search mechanism, firms produce to meet demand, paying a wage determined by their own labor demand schedule but hiring only enough workers to meet their expected demand in each period. This effectively removes the labor market equilibration from Farmer's otherwise conventional RBC-based model and renders the steady state of the model indeterminate. Farmer closes the model by introducing a belief function by which agents forecast future wealth, which then determines demand. The indeterminacy in the model means that any belief function equilibrium will constitute a REE. The arbitrariness of the belief function then allows Farmer (and others working in similar frameworks) to generate equilibrium time-series that match the data much more closely than conventional DSGE models.

Farmer's approach has not gained much traction in macroeconomics, likely for a number of reasons. First, the lack of theoretical underpinnings for expectation formation has been a critique of economic models exhibiting indeterminacy from Jean Michel Grandmont's early work on temporary equilibrium models, to models of sunspot equilibria (self-fulfilling rational expectations equilibria), and to the current work on models of Keynesian search. This arbitrariness of expectations can be problematic for imposing what amount to exogenous constraints on the operation of the financial and product markets, which may or may not be valid.

Second, the fact that expectations can be arbitrary reflects a fundamental lack of knowledge about psycho-economic factors that determine how consumer expectations and sentiment are generated and which might discipline the specification of expectations. This situation is starting to change, with work being done in economic theory on Bayesian persuasion games, and other work (generally being done in the context of agent-based modeling) using network dynamics of rumor propagation to construct models of sentiment formation (see, e.g., Gomes (2015)). Farmer himself has addressed this issue in a number of papers; see Benhabib and Farmer (1999).

Although it is by now well-established that the equilibria in general SOLG models are generically history-dependent, the mechanism that gives rise to this result may not be familiar to some readers. Since this is also a fundamental difference between standard DSGE and SOLG models, it is worth unpacking this result in some detail. In the SOLG framework, portfolio rebalancing in the face of shock realizations makes the competitive equilibrium history dependent. (We will give a demonstration of this result in the context of the three-period Diamond model below.) Woodford (1986) showed how to solve these models using the infinite history of shocks as the state variable. Later work by Spear and Srivastava (1986), Duffie et al. (1994) and Ríos-Rull (1996) showed how to use lagged endogenous variables as sufficient statistics for the shock history. Rios-Rull's assumption that the wealth distribution would be the natural statistic was subsequently shown to be true generically by Citanna and Siconolfi (2010). It is, of course, true that standard DSGE models include lagged endogenous state variables – the capital stock is the most obvious – but these do not serve as sufficient statistics for shock histories in these models. Indeed, in the RBC setting, the capital stock is determined by the law of motion for capital together with the optimal choice of saving by households. It is also true that in the standard DSGE setting, portfolio rebalancing decisions generate income effects, but these effects are internalized by the infinite lived households in the model through adjustments to bequests in a way that keeps the intertemporal allocations *ex ante* optimal. This is the feature that is missing in the SOLG setting.

The approach we propose here is one of mapping specified production processes into REE equilibrium in the product and financial markets. This pins down the expectations of agents about what their wealth must be and, as we will show, determines an explicit expectations mapping. The value in this approach stems from the fact that while we don't have good empirical models of expectation formation, we have many carefully done models of the empirical stochastic process of aggregate production that we can use to calibrate models with recursive equilibria. This leaves the determination of the distribution of output and financial assets endogenous and provides a mechanism by which we can compare the operation of these markets across different macroeconomic modeling approaches to answer questions of model validity and possible observational equivalences. The approach leaves unanswered the question of how expectations might be formed, but it also provides a platform for evaluating how well different expectation formation mechanisms fit with the equilibrium outcomes from simulations. However, we emphasize that the approach we are taking here is agnostic on the causality question of whether it is expectations that generate outcomes or vice versa. We agree with Farmer's view that, in fact, the causality runs from expectations to equilibrium outcomes, though in stationary rational expectations models these are obviously determined simultaneously.

We use the model developed here to examine a number of issues. The first concerns the disparity in hours worked by age. Ríos-Rull (1996) used a multi-period SOLG model to examine how life cycle differences affected a number of macroeconomic variables, among them the hours worked by agents of different ages. He found that the model (which includes both aggregate and idiosyncratic shocks) matched the observed hours worked of older agents but could not replicate the hours of younger workers. This discrepancy suggests that while conventional labor market models (whether classical or New Keynesian) do an adequate job of explaining employment of older workers, a new approach is needed to understand what is going on at the entry level. This observation then suggests that lifecycle considerations may be important for the emergence of dual labor markets consisting of a primary sector employing skilled, experienced workers and a secondary sector consisting of entry-level workers with "low skill levels, low earnings, easy entry, job impermanence, and low returns to education or experience" (Beer and Barringer (1970)).

We model the dual labor market structure by working with a multi-period SOLG model in which older, seasoned agents act as managers who, among other activities, participate in the recruiting of new younger agents via the Keynesian search process from Farmer's work [2]. This lifecycle interpretation is well grounded in empirical work showing that young workers optimally sample different jobs/occupations, which generates a significant degree of employment churn, before settling into long-duration spells of employment with specific firms or within specific industries. (See Miller (1984) for early work on this or Sullivan (2010) and references therein for a more recent survey of this work.) The model developed here then allows us to use hours worked by the unseasoned younger workers as a calibrated parameter that drives the stochastic fluctuations of the model. We then show how to use standard results from pure exchange SOLG models or SOLG models with capital accumulation (and otherwise complete markets) to determine the REE of the dual labor markets model, which, following Farmer, we then interpret as the belief function of firms that gives rise to the observed variation in hours worked by the young in equilibrium.

This lifecycle interpretation also connects the model to recent work on labor-market tightness and Beveridge-curve shifts. Recent research treats movements in unemployment and vacancies not only as movements along a stable matching relation, but also as shifts generated by changes in matching efficiency, worker search behavior, reallocation, and labor-force participation. In our model, shocks to recruiting productivity, which we use in place of shocks to TFP, play this reduced-form role: they shift the mapping between seasoned labor allocated to recruiting and employment of entry-level workers, thereby affecting output, asset prices, and the endogenous wealth distribution. This interpretation is consistent with recent evidence on Beveridge-curve shifts and post-pandemic labor-market tightness (Barlevy et al., 2024; Kindberg-Hanlon and Girard, 2024; Anderson and Lubik, 2026). Our reduced-form shock is also consistent with evidence that worker search behavior itself varies systematically over the cycle: search effort is countercyclical (Mukoyama et al., 2018), and on-the-job search alone can generate cyclical movements in the Beveridge curve (Eeckhout and Lindenlaub, 2019). This evidence points to variation in the effectiveness with which workers and vacancies are matched as a central driver of employment fluctuations — precisely the channel our shocks to recruiting productivity are designed to capture.

In computational terms, our approach is also related to a series of papers which examine the effects of aggregate shocks on labor market dynamics in Aiyagari-Bewley-

[2] Firms' decisions on how much seasoned labor to allocate to search generates search intensity effects similar to those originally analyzed by Howitt and McAfee (1987).

Huggett models, which feature incomplete asset markets and household heterogeneity (see Krusell et al. (2010) for the canonical combination of precautionary savings and search-and-matching frictions in an aggregate model with incomplete markets). Our SOLG framework can be seen as replacing the infinitely-lived, incomplete-market household in that literature with finitely-lived overlapping generations households, which sharpens the discipline that the wealth distribution imposes on admissible belief functions.

A second, more general set of issues that we can address in the model concern the relationship between employment uncertainty and employment across the business cycle. For this analysis, our assumption that shocks to matching generate changes in employment for the young amounts to the imposition of a form of employment uncertainty in the matching process. This assumption is supported by a substantial literature on the empirical importance of uncertainty in labor markets — see, for example, Barnichon and Figura (2010) or Ravn and Sterk (2017). As Barnichon and Figura note, the existence of shocks to the matching function provides a mechanism to explain the apparent shifts in the Beveridge curve over time, and their estimates of these shocks provide one benchmark for calibrating our model. Other work emphasizes that changes in measured matching efficiency may instead reflect shifts in the composition of job seekers and greater segmentation across labor markets: Barnichon and Figura (2015) show that these factors account for an important part of the decline in measured matching efficiency after 2007, and, after adjusting for heterogeneity among job seekers, Hall and Schulhofer-Wohl (2018) similarly find that much of the apparent deterioration during the Great Recession reflects compositional change. Coles and Moghaddasi Kelishomi (2018), in turn, show that job-destruction shocks can generate the joint movement of unemployment and vacancies associated with Beveridge-curve dynamics when vacancy creation is not perfectly elastic. Together, these findings support treating matching efficiency as time varying, while also suggesting that measured shifts need not arise solely from changes in recruiting technology.

Given this calibration, we can then determine the stochastic process of equilibrium in the model, conditional on the state of the business cycle, and ask whether the simulations match up with the data. In the quantitative version of the model where we calibrate production using the results from regime-switching econometric models of the business cycle, we can examine the extent to which time homogeneous shocks to matching are consistent with the nonlinearity of the regime-switching process. Conversely, given data on the movement of the Beveridge curve over the business cycle, we can calibrate time-varying shocks to match and then examine whether the resulting stochastic process of employment is consistent with the data. Finally, we note that it is entirely possible to interpret the shocks to matching as extrinsic uncertainty, i.e., sunspots.

Empirical examples of this include the apparent discrimination against the long-term unemployed during the recovery from the recession induced by the 2008 financial crisis (see Kroft et al. (2016)). A second example is the so-called "Big Quit" following the covid-19 pandemic lockdowns, in which large numbers of laid off workers either exited the work force or spent above average amounts of time searching for new jobs. The most recent data on work force participation strongly suggests that this was a transient phenomenon. We view modeling these actions as random shocks as a way of cutting through the need to formally model the psychological or policy-driven motives behind these examples.

A third set of issues we can address involve the product and financial market side of the model. Once we specify how wages and profits are determined on the production side, the REE of the associated exchange economy will then generate expectations (possibly over sunspots or sentiment variables) such that demand is compatible with production. The resulting REE will generate time series data in simulations for the equilibrium stochastic process on output, asset prices and consumption, which lets us examine how well these markets match up with empirical data over the course of the business cycle.

SOLG models are, of course, not the only contemporary macroeconomic models to feature heterogeneity, and related models like the Heterogeneous Agent New Keynesian (HANK) model are also being widely studied. These models place the distribution of income, wealth, and liquid assets at the center of aggregate shock transmission. In HANK models, the cross-sectional distribution affects marginal propensities to consume, precautionary saving, portfolio choices, and the general-equilibrium response of labor income and asset prices to aggregate shocks. At the same time, Keynesian-search models following Farmer emphasize that when wages do not clear the labor market, beliefs about future income and asset values can become an equilibrium object rather than a passive forecast.

SOLG and HANK models intersect in requiring that the rational expectations equilibrium of the model be a fixed point between forecasts and temporary equilibrium outcomes. This connection can be seen clearly in the heterogeneous agent computational literature (Kaplan et al., 2018; Auclert et al., 2021; Bayer et al., 2024). The models differ in their assumptions about the labor market, and we note that the HANK framework could be applied under Farmer's assumption of labor market incompleteness. It would be interesting to compare the results from such a model with those arising in the SOLG setting.

There is one important difference between HANK and SOLG models, however. HANK models cannot capture the longitudinal heterogeneities associated with the human lifecycle, while SOLG models can (and have) incorporated cross-sectional heterogeneities.[3] Nevertheless, despite this difference in scope, the discipline we uncover – that belief rules must reproduce the endogenous wealth distribution they presuppose – is a structural feature of heterogeneous-agent equilibrium more broadly, including HANK-style.

In Section 2 of the paper, we develop our formal model in the relatively simple context of three-period lived agents. This makes the otherwise complex interactions in the model reasonably transparent and serves to illustrate the key differences between the SOLG and standard DSGE approaches. In this section we also develop the general model in which agents live 58 periods after entering the market at age 24, under the assumption that markets are perfectly competitive. This model is the basis for our simulations. To efficiently solve this complex, high-dimensional lifecycle model, we employ a neural network-based numerical approach inspired by recent advances in computational methods for overlapping generations frameworks (Azinovic and Zemlicka (2024)). Specifically, this method iteratively trains neural networks (NN) to approximate equilibrium policy functions by minimizing Euler equation errors related to agents' optimal capital and bond accumulation decisions, alongside errors in bond price determination and constraints enforcing consumption feasibility. The flexibility and computational efficiency of neural networks enable us to accurately handle substantial agent heterogeneity and extensive state spaces, significantly reducing computational complexity compared to traditional numerical methods. Section 3 describes the calibration for both the illustrative three-period model and the baseline 58-period economy. Section 3.2 presents the neural-network solution method and the simulation algorithm. Section 3.3 reports quantitative results from the calibrated 58-period economy. Section 3.4 backs out and interprets the implied expectations function consistent with the recursive rational-expectations equilibrium. Section 4 concludes.

2. Model

2.1. Three-period Framework

To illustrate the basic workings of the model, we first examine a three-period version of it, since this keeps the notation relatively simple. Our simulations will be based on a

[3] See Ríos-Rull (1996) and the many papers inspired by this for examples.

more general model in which households live for 58 periods, entering the market at age 24 and dying at age 81. The model is a stochastic version of the Diamond model, but following Farmer (2012, 2013), we assume that the labor market for new entrants (i.e., young agents) is incomplete.

This labor-market friction generates a Farmer-style steady-state indeterminacy. In a standard Diamond model, the labor-market clearing condition helps pin down stationary employment, output, capital, and prices. Here, the entry-level wage does not perform that equilibrating role. A stationary allocation therefore requires an additional equilibrium restriction, such as a belief rule, an employment rule, or a bargaining rule. Below, we develop this point in a deterministic Diamond-Farmer benchmark and then use a two-state stochastic three-period example to show why a current-shock-only belief rule generally cannot close the model. The stochastic example motivates the recursive equilibrium steady state of our main models, in which the state variables are the current aggregate shock together with lagged endogenous asset holdings, or equivalently the cross-sectional wealth distribution.

The stochastic three-period economy clarifies why an arbitrary belief rule cannot serve as a closure device in the SOLG framework. Labor-market incompleteness removes the usual wage-clearing restriction, just as in the deterministic benchmark above. In that deterministic setting, this restriction can be supplied directly by a belief rule, an employment rule, or a bargaining rule, and any of these choices closes the model. In the stochastic SOLG setting, however, rational expectations requires the induced cross-sectional distribution of asset holdings to coincide with the distribution agents use when forming forecasts. This consistency requirement disciplines which belief rules are admissible, but it does not by itself supply a closure device. What it shows is that any admissible belief rule must condition on the lagged wealth distribution — the state-space enlargement on which our recursive equilibrium concept, and ultimately our calibration, is built.

The stochastic OLG environment also introduces a standard asset-market issue. Households trade capital claims and one-period bonds. These assets need not span all date-event claims, so asset markets may be incomplete in the Arrow-Debreu sense. This form of asset-market incompleteness is distinct from labor-market incompleteness. Labor-market incompleteness creates the Farmer-style steady-state indeterminacy; asset-market incompleteness, finite lives, and portfolio rebalancing make the wealth distribution an endogenous state variable in recursive equilibrium. We will show below in the context of a simple example that the need to consider history dependent equilibria in SOLG models is not the result of asset market incompleteness.

In the absence of an organized market for entry-level inputs, firms in the model must engage in costly search to determine the suitability, numbers hired, and wages paid to young workers. We model this via the search process outlined in Farmer (2012, 2013). As outlined in the introduction, young workers who pass the search screen face a take-it-or-leave-it decision on whether to accept employment at the wage offered by the firm. This wage can be determined in a number of different ways. Farmer (2012, 2013) assumes that firms pay competitive wages once the screened pool of applicants has been determined. We will follow Farmer and assume that wages are determined competitively as the marginal product of employed young workers. We emphasize, however, that wages do not clear the labor market for such unseasoned labor: employment in any period in the model is determined by firms' expectations of current demand. To simplify the exposition, we also assume that workers supply their labor inelastically (with respect to wages). Other than this assumption on wage determination, the search process in the model is standard. The costs of search in the model take the form of diverting seasoned (i.e. middle-aged) workers from production to search.

The seasoned-labor market clears competitively. Middle-aged workers supply labor inelastically, and the seasoned wage adjusts to equate the supply and demand for seasoned labor. The empirical justification for this is based on the observed differences between the volatility of hours worked by middle-aged agents relative to young agents, together with Miller (1984)'s findings on the relationship between age and job stability. As noted above, firms must allocate seasoned workers to either production activities or search activities, which requires that firms forecast the demand they expect to face and then allocate search resources to hire entry-level workers who engage in production with other seasoned workers allocated to production.

Let N_1 and N_2 denote the number of young and middle-aged workers employed in production, with K the amount of capital employed. The representative firm's production function is then given by

$$Y = F\left(N_1, N_2, K\right) \tag{1}$$

where F is C^2 and concave. For specificity, we work with a Cobb-Douglas formulation of production for our examples and simulations, so that

$$Y = N_1^{\alpha_1} N_2^{\alpha_2} K^{\alpha_k} \tag{2}$$

Following standard practice, we will work in per capita terms, though it will simplify our exposition to be able to work with the two types of labor as being relative to the total amount of each type employed. So, we write

$$Y = \left(\frac{L_1}{L_1} N_1\right)^{\alpha_1} \left(\frac{L_2}{L_2} N_2\right)^{\alpha_2} \left(\frac{L_1+L_2}{L_1+L_2} K\right)^{\alpha_k} = L_1^{\alpha_1} L_2^{\alpha_2} \left(L_1+L_2\right)^{\alpha_k} n_1^{\alpha_1} n_2^{\alpha_2} k^{\alpha_k} \tag{3}$$

where L_i is the population of type $= 1, 2$,and

$$n_1 = \frac{N_1}{L_1} \tag{4}$$

$$n_2 = \frac{N_2}{L_2} \tag{5}$$

$$k = \frac{K}{L_1+L_2} \tag{6}$$

Dividing through by $L_1 + L_2$ and defining

$$y = \frac{Y}{L_1+L_2} \tag{7}$$

equation (3) yields

$$y = B n_1^{\alpha_1} n_2^{\alpha_2} k^{\alpha_k} \tag{8}$$

where $B = \frac{L_1^{\alpha_1} L_2^{\alpha_2}}{(L_1+L_2)^{1-\alpha_k}}$. For the general production function, we write $y = f\left(n_1, n_2, k\right)$ to denote output in per capita terms. As long as population doesn't change and the relative populations of the two types of labor remains the same, B will be constant. For our analysis of competitive markets, we will assume that the production technology exhibits constant returns to scale, so that $\alpha_1 + \alpha_2 + \alpha_k = 1$. As noted above, we follow Farmer in assuming that firms are myopic and act to maximize profit in the current period. More complicated versions of the model would relax this assumption. For an interesting model of more general forward-looking behavior in a model of optimal contracting with moral hazard, see Li and Wang (2022).

We assume that there are a continuum of households of each age (with the measure of each normalized to 1). Consumption is the numeraire good, and agents in the model can save by purchasing capital assets issued by the firm or by trading bonds in zero net supply. Firms can convert each unit of capital assets sold into a unit of productive capital. We denote bond quantities sold at time t by b_t and assume they sell at time t at a price $\phi_t < 1$ and are paid off the following period at par. Firms' assets k_t are in terms of the final good, i.e. they are sold at a price of 1, the firms pay a gross return of $r_t + (1-\delta) > 1$ where δ is the depreciation rate.

Agents are endowed with 1 unit of possible labor input and do not value this directly. Young agents just entering the labor market must first spend time searching for work and, if hired, they supply a single unit of endowed labor time inelastically. To keep the model tractable we follow the usual approach and assume that every household ends up supplying some fraction $0 \leq n_1 \leq 1$ of their available labor depending on the employment level determined by the search process. For middle-aged agents, labor supply is denoted by n_2. Because the seasoned-labor market clears competitively and middle-aged workers supply labor inelastically, equilibrium seasoned employment is $n_2 = 1$.

A typical young agent then faces the following consumption optimization problem (omitting time subscripts for simplicity)

$$\max_{\{c_1,c_2,c_3\}} \; u(c_1) + \beta \mathbb{E}\left[u(c_2)\right] + \beta^2 \mathbb{E}\left[u(c_3)\right] \tag{9}$$

subject to budget constraints of the form

$$c_1 = w_1 n_1 - \phi b_1 - k_1 \tag{10}$$
$$c_2 = w_2 n_2 + b_1 - \phi' b_2 + k_1(r + 1 - \delta) - k_2 \tag{11}$$
$$c_3 = b_2 + k_2(r' + 1 - \delta) \tag{12}$$

where ϕ and ϕ' are the bond prices in the first and second periods, and r and r' are the rental rates on capital in the two periods. Old agents do not work and must finance their retirement consumption from the returns on their asset holdings.

The asset and bond market clearing conditions in this economy are

$$b_1 + b_2 = 0 \tag{13}$$
$$k_1 + k_2 = k \tag{14}$$

where k is the current capital stock. Walras' law ensures that when the bond and capital markets clear, the market for consumption will also clear.

For reference, it is useful to first write the household optimality conditions in the deterministic perfect-foresight case. For an interior optimum, the first-order conditions with respect to b_1, k_1, b_2, and k_2 are

$$\phi u'(c_1) = \beta u'(c_2), \tag{15}$$
$$u'(c_1) = \beta(1 + r - \delta)u'(c_2), \tag{16}$$
$$\phi' u'(c_2) = \beta u'(c_3), \tag{17}$$
$$u'(c_2) = \beta(1 + r' - \delta)u'(c_3). \tag{18}$$

Thus, when both assets are held in positive amounts and returns are riskless, the bond and capital Euler equations imply the no-arbitrage restrictions

$$\frac{1}{\phi} = 1 + r - \delta, \qquad \frac{1}{\phi'} = 1 + r' - \delta.$$

A deterministic perfect-foresight equilibrium of the three-period economy consists of allocations $\{c_1, c_2, c_3, b_1, b_2, k_1, k_2, n_1, n_2, k\}$ and prices $\{w_1, w_2, r, r', \phi, \phi'\}$ such that households solve the lifetime problem, firms solve the static production and recruiting problem, the seasoned-labor market clears competitively, the entry-level labor input is generated by the recruiting technology, and bond, capital, and goods markets clear.

Three notions of indeterminacy should be distinguished. The first is steady-state indeterminacy. In the deterministic Diamond-Farmer benchmark, removing the wage-clearing condition from the entry-level labor market removes one equilibrium restriction, leaving a continuum of stationary allocations unless an additional behavioral relation closes the model. We examine this indeterminacy in detail below.

The second is dynamic indeterminacy: for a given stationary environment, there may be multiple equilibrium transition paths or belief-driven laws of motion around a steady state. Farmer (2020) emphasizes this distinction. Our quantitative exercise does not select an arbitrary dynamic path. It computes a stationary recursive rational-expectations equilibrium in which the forecast rule, temporary equilibrium allocations, and induced wealth distribution are mutually consistent.

The third is the classic monetary indeterminacy of OLG models. That is not the mechanism studied here. The benchmark economy has no fiat money, and the indeterminacy we analyze is not a monetary OLG indeterminacy. It comes from the absence of a wage-clearing condition in the entry-level labor market.

In the market-clearing Diamond benchmark, the usual dynamic-efficiency condition can be checked by comparing the gross return on capital with the growth rate of the economy. The indeterminacy analyzed here does not rely on over accumulation or on the coexistence of monetary and non-monetary OLG steady states.

2.2. Farmer Indeterminacy in a Deterministic Diamond Model

To make clear what we are doing, we illustrate the steady state indeterminacy in the simple Diamond model under the additional simplification that all agents (both young and middle-aged) face incomplete labor markets. This simplification makes the example parsimonious by omitting the firms' matching problems, the inclusion of which doesn't change the basic indeterminacy result.

Households. As above, agents live three periods, denoted as $1, 2, 3$ for young, middle-aged and retired. The young and middle-aged offer labor $n \in [0, 1]$ to firms who produce a single output Y using a Cobb-Douglas technology taking inputs of capital and labor

$$Y = AK^{\alpha}L^{1-\alpha}.$$

Since the production function is constant-returns-to-scale, we can work in per capita terms and write

$$y = Ak^{\alpha}l^{1-\alpha}$$

where the lower case letters denote per capita terms. Retired agents do not work and must finance consumption from savings. Households save by purchasing capital which they then rent to firms for production. Since output and capital are the same, and we take output as the numeraire, the price of output and capital will be one. Budget constraints at a steady state for a typical household over its lifetime will be

$$\begin{aligned} c_1 &= wn - k_1 \\ c_2 &= wn + rk_1 - k_2 \\ c_3 &= rk_2 \end{aligned}$$

where r is the rental rate for capital and w is the wage. We don't include the bond because multiple assets in a deterministic setting are redundant.[4]

Households have preferences over lifetime consumption of the form

$$U(c_1, c_2, c_3) = ln(c_1) + \beta ln(c_2) + \beta^2 ln(c_3)$$

where $\beta \in [0, 1]$ is the household's discount factor.

The first-order conditions for the household's intertemporal utility maximization with respect to k_1 and k_2 are

[4]For this section, we assume that depreciation is 100% so that there is no difference between gross and net rates of return.

$$-\frac{1}{wn-k_1}+\frac{\beta r}{wn+rk_1-k_2}=0$$
$$-\frac{1}{wn+rk_1-k_2}+\frac{\beta}{k_2}=0.$$

These equations can be solved for the optimal steady state capital holdings of each household. Note that because the households incur no disutility from labor, while they take the wage as given, the optimal labor offer in each period is $n=1$. Note, however, that equilibrium employment will leave n indeterminate.

Firms. Firms are perfectly competitive and act to maximize profits

$$\pi = y - wl - rk$$
$$= Ak^{\alpha}l^{1-\alpha} - wl - rk$$

where (as above) r is the rental rate for capital, and w is the wage.

The first-order condition for the firm's profit maximization problem are

$$\alpha Ak^{\alpha-1}l^{1-\alpha} - r = 0$$
$$(1-\alpha)\,Ak^{\alpha}l^{-\alpha} - w = 0.$$

These can be put into the usual form

$$\alpha\frac{y}{k} = r$$
$$(1-\alpha)\,\frac{y}{l} = w.$$

Equilibrium. In a standard Diamond model with complete capital and labor markets, the competitive equilibrium would be determined by the capital and labor market clearing conditions

$$l = 2n$$
$$k = k_1 + k_2.$$

In this setting, since households incur no disutility from labor, the labor market clearing condition would be simply
$$l = 2.$$

The first-order conditions for households and firms, together with the market clearing equations (taking Walras' law into account) can then be solved for the equilibrium wage w^* and capital rental rate r^* in the usual way.

In Farmer's incomplete labor markets framework, the wage plays no role in clearing the labor market and can be fixed arbitrarily. In this case, given the wage w, the firm will hire labor sufficient to meet it's expected demand. From the FOCs for profit maximization, this implies that
$$l = (1-\alpha)\,\frac{y}{w}$$
so that household employment will be

$$n^* = \frac{(1-\alpha)\,y}{2w}.$$

Similarly, given its expected demand, the firm will purchase capital such that

$$k^* = \alpha\frac{y}{r}.$$

The labor market incompleteness then yields an indeterminate steady state for the model. To see this, we solve the household FOCs for their optimal capital holdings (taking wn as given). To simplify the calculation, we take $\beta = 1$. The first FOC implies that

$$\begin{aligned} r\left[wn - k_1\right] &= wn + rk_1 - k_2 \\ \Rightarrow (r-1)\, wn &= 2rk_1 - k_2. \end{aligned}$$

The second FOC implies that

$$\begin{aligned} k_2 &= wn + rk_1 - k_2 \\ \Rightarrow 2k_2 &= wn + rk_1. \end{aligned}$$

Solving these simultaneously for k_1 and k_2 yields

$$\begin{aligned} \hat{k}_1 &= \frac{2r-1}{3r} wn \\ \hat{k}_2 &= \frac{1+r}{3} wn. \end{aligned}$$

In equilibrium, we require that

$$\begin{aligned} \hat{k}_1 + \hat{k}_2 &= k^* \\ \Rightarrow \alpha \frac{y}{r} &= \frac{1}{3r}\left[r^2 + 3r - 1\right] wn \\ \Rightarrow \alpha y &= \frac{1}{6}\left[r^2 + 3r - 1\right](1-\alpha)\, y. \end{aligned}$$

The level of output y here is clearly indeterminate. The equilibrium equation will determine the steady state interest rate via the solution to the quadratic in r, but the allocations of capital and consumption depend on the employment levels, which depend on y. We can close the model by specifying a steady state expectations function for households and firms in terms of employment n and the resulting equilibrium will be a fully rational expectations equilibrium. This is fully in parallel with Farmer's construction in the conventional DSGE framework.

2.3. Stochastic Extension of the Diamond Model Dynamics

Since many readers may not be familiar with the aspects of general SOLG models that give rise to history dependence of the competitive equilibrium, we outline the derivation of these results here in the context of the simplified Diamond model used above to illustrate the Farmer indeterminacy results. In this simple model, we will assume that there are two stochastic shocks to employment and two assets available – capital and zero net supply bonds. This makes the asset market structure in the model sequentially complete.

Unlike in the DSGE setting, where this specification of shocks would lead to an equilibrium in which the endogenous variables are simple functions of the shocks, in the three period SOLG environment, the competitive equilibrium will depend on the whole history of the shock process at a stochastic steady state equilibrium. For our analysis of this case, then, we set

$$n_t = n_t^s,\ s = H, L$$

where the states H or L occur with probability $1 > \Pi^s > 0$. With this assumption, firms will produce output

$$y^s = A\left[k^s\right]^{\alpha}\left[n^s\right]^{1-\alpha}$$

where we have thrown the additional constant factor $2^{1-\alpha}$ into the leading coefficient. We also assume that the aggregate capital stock depends only on the current shock s.

Note that one can extend the analysis we present below to the case of infinitely many shocks using the fact that any Borel probability measure can be approximated as the limit of a sequence of finite measures with discrete support.

We now define the notions of equilibrium we use.

Definition: A *strongly stationary* equilibrium is one in which all endogenous variables only depend on the current period exogenous shock. A *short memory* equilibrium is one in which all endogenous variables depend on a fixed finite number of lagged exogenous shocks.

We will consider the strongly stationary case first.

To specify household behavior in this version of the model, we will introduce a zero net supply bond that households can hold in addition to capital in order to demonstrate that the history dependence result does not turn on whether or not the asset markets are sequentially complete. With the addition of the bond, households now face two shocks but have two assets with which to transfer wealth over time and across states, so markets are sequentially complete.

With this addition, the optimization problem facing a typical young agent born into state s will be

$$\max_{\left(c_1^s,\, c_2^{ss'},\, c_3^{s's''}\right)} u\left(c_1^s\right) + \beta \sum_{s' \in \{H,L\}} \Pi^{s'} u\left(c_2^{ss'}\right) + \beta^2 \sum_{s' \in \{H,L\}} \sum_{s'' \in \{H,L\}} \Pi^{s'} \Pi^{s''} u\left(c_3^{s's''}\right).$$

subject to

$$\begin{aligned} c_1^s &= wn^s - \phi^s b_1^s - k_1^s \\ c_2^{ss'} &= wn^{s'} + b_1^s + r^{s'} k_1^s - \phi^{s'} b_2^{s'} - k_2^{s'} \\ c_3^{s's"} &= b_2^{s'} + r^{s"} k_2^{s'} \end{aligned}$$

where ϕ^s is the price of the bond in state s and r^s is the rental rate for capital in state s. Note that we maintain the assumption that the wage is fixed since it plays no role in market clearing. A key thing to note about the budget constraints is that we allow explicitly for consumption in middle-age and retired periods to depend on the realization of the previous period's state because bond and capital holdings in these budget constraints depend on this. For the assumption that equilibrium consumptions only depend on the current shock to hold, we will need to show that prices are such that agents will choose the same consumptions in any period regardless of the history of shocks.

The young household's Euler equations for the optimization take the form

$$\begin{aligned} \phi^s u'\left(c_1^s\right) &= \beta\left[\Pi^H u'\left(c_2^{sH}\right) + \Pi^L u'\left(c_2^{sL}\right)\right] \\ u'\left(c_1^s\right) &= \beta\left[r^H \Pi^H u'\left(c_2^{sH}\right) + r^L \Pi^L u'\left(c_2^{sL}\right)\right] \end{aligned}$$

$$\begin{aligned} \phi^{s'} u'\left(c_2^{ss'}\right) &= \beta\left[\Pi^H u'\left(c_3^{s'H}\right) + \Pi^L u'\left(c_3^{s'L}\right)\right] \\ u'\left(c_2^{ss'}\right) &= \beta\left[r^H \Pi^H u'\left(c_3^{s'H}\right) + r^L \Pi^L u'\left(c_3^{s'L}\right)\right] \end{aligned}$$

for any $s = H, L$. When we substitute for the consumptions from the budget constraints, the market clearing conditions will be

$$\begin{aligned} b_1^s + b_2^s = 0 \qquad & s = H, L \\ c_1^s + c_2^{s^{-1}s} + c_3^{s^{-1}s} + k^s = y^s \qquad & \left(s^{-1}, s\right) \in \{H.L\}^2 \end{aligned}$$

where s^{-1} refers to the shock realized one period past, k^s is the aggregate capital stock and y^s is as above.

We don't include a clearing condition on the capital market since $k^s = k_1^s + k_2^s$ determines the capital stock. From these equilibrium conditions, we note first that from the second set of marginal conditions, we have that $c_2^{ss'}$ cannot depend on s, since the right-hand term doesn't depend on this lagged state. Since second period consumption can only depend on the current state, and first period consumption is only dependent on the current state, with k and y dependent only on the current state, it follows from the last market clearing equation that third period consumption can only depend on the current shock. Hence, everything in the model is now consistent in that in equilibrium, all endogenous variables are functions of the current shock alone. We can now show the following.

Proposition: There cannot exist a strongly stationary rational expectations equilibrium.

Proof: We begin by eliminating the product market clearing condition via Walras' Law. Adding the budget constraints yields

$$\begin{aligned} c_1^s + c_2^{s^{-1}s} + c_3^{s^{-1}s} &= wn^s - \phi^s b_1^s - k_1^s \\ &+ b_1^{s^{-1}} + r^s k_1^{s^{-1}} - \phi^s b_2^s - k_2^s \\ &+ b_2^{s^{-1}} + r^s k_2^{s^{-1}} \end{aligned}$$

Collecting terms with like coefficients yields

$$\begin{aligned} c_1^s + c_2^{s^{-1}s} + c_3^{s^{-1}s} &= 2wn^s + \left[b_1^{s^{-1}} + b_2^{s^{-1}}\right] \\ &- \phi^s \left[b_1^s + b_2^s\right] + r^s \left[k_1^{s^{-1}} + k_2^{s^{-1}}\right] - \left[k_1^s + k_2^s\right] \\ &= 2w^s n^s + r^s k^{s^{-1}} - k^s \end{aligned}$$

where we use the bond market clearing condition and the fact that the sum of individual capital holdings defines the aggregate capital stock in the last step. Note that the factor share relationships for the Cobb-Douglas production function (or any constant returns to scale production function) will imply that

$$c_1^s + c_2^{s^{-1}s} + c_3^{s^{-1}s} + k^s = 2wn^s + r^s k^{s^{-1}} = y^s$$

which is just the product market clearing condition. Walras' law follows from this. Given this result, we can drop the product market clearing equations and Walras' Law conditions to obtain a system of 10 equations – 4 Euler equations for the young and middle-aged respectively, for a total of 8 equations, plus the two bond market clearing equations. We also have ten variables – two bond holdings for young and middle age, two capital holdings for young and middle-aged, plus two bond prices – but not all of these are independent. To show this, we use the fact that the consumption of the middle-aged agents cannot depend on history. From the budget constraints, we have

$$\begin{aligned} c_2^{HH} &= wn^H + b_1^H + r^H k_1^H - \phi^H b_2^H - k_2^H \\ c_2^{LH} &= wn^H + b_1^L + r^H k_1^L - \phi^H b_2^H - k_2^H. \end{aligned}$$

Subtracting the second equation from the first yields

$$b_1^H + r^H k_1^H = b_1^L + r^H k_1^L.$$

Similarly, the requirement that $c_2^{LL} = c_2^{HL}$ implies that

$$b_1^H + r^L k_1^H = b_1^L + r^L k_1^L.$$

Subtracting the second equation from the first yields

$$\left[r^H - r^L\right] k_1^H = \left[r^H - r^L\right] k_1^L.$$

This gives us two possible cases: either $r^H = r^L$ or $k_1^H = k_1^L$. The first condition is ruled out by the fact that r^s is determined by the firms' marginal products of capital; if these are constant, then employment must also be constant, which is a contradiction. Hence, we are left with the implication that $k_1^H = k_1^L$. The requirement that the consumption of retired households also be independent of histories then also implies that $k_2^H = k_2^L$. Plugging the bond holding implications back into the conditions above then also implies that $b_1^H = b_1^L$. The bond market clearing equations will then imply that the bond holdings of the middle-aged households are also independent of the exogenous shock state. With these restrictions, then, we are left with only 6 free variables: b_1, b_2, k_1, k_2, ϕ^H, ϕ^L. [5] Thus the current-shock Markov candidate is generically overdetermined: it imposes more independent equilibrium restrictions than there are free variables. The conclusion is not that the model is closed by the current shock, but rather that the state space must be enlarged to include lagged endogenous variables, such as the distribution of asset holdings. To complete the proof and show that non-existence of this equilibrium is generic, one can use the perturbation methods found in Citanna and Siconolfi (2010), Orrego (2026), or Spear (1985). □

As a corollary to this proposition, we show that there cannot be a short memory REE in the model. By way of notation, we fix some finite $k > 1$ and let $\sigma_k = [s_{-k}, s_{-k+1}, ..., s_{-1}]$ be the finite sequence of lagged shocks the endogenous variables can depend on. It will be convenient in what follows to let

$$\sigma_k = [s_{-k}, \sigma_{k-1}]$$

Proposition: There cannot exist a short memory rational expectations equilibrium.

Proof:

The Euler equation for the bond holdings of a middle-aged agent now take the form

$$\phi^{\sigma_{k-1}s} u' \left[c_2^{\sigma_k s}\right] = \beta \left[\Pi^H u'\left(c_3^{\sigma_{k-1}sH}\right) + \Pi^L u'\left(c_3^{\sigma_{k-1}sL}\right)\right].$$

It follows from this that with

$$c_2^{\sigma_k s} = c_2^{s_{-k}\sigma_{k-1}s}$$

second period consumption cannot depend on s_k since the right-hand side of the Euler equation doesn't depend on s_k.

Parallel to the calculation above for the strongly stationary carse, this implies that

$$c_2^{H\sigma_{k-1}H} = wn^H + b_1^{H\sigma_{k-1}} + r^{\sigma_{k-1}H} k_1^{H\sigma_{k-1}} - \phi^{\sigma_{k-1}H} b_2^{\sigma_{k-1}H} - k_2^{\sigma_{k-1}H}$$

and

$$c_2^{L\sigma_{k-1}H} = wn^H + b_1^{L\sigma_{k-1}} + r^{\sigma_{k-1}H} k_1^{L\sigma_{k-1}} - \phi^{\sigma_{k-1}H} b_2^{\sigma_{k-1}H} - k_2^{\sigma_{k-1}H}.$$

Subtracting the second equation from the first taking account of the fact that the second period consumptions must be the same, we get

$$b_1^{H\sigma_{k-1}} + r^{\sigma_{k-1}H} k_1^{H\sigma_{k-1}} = b_1^{L\sigma_{k-1}} + r^{\sigma_{k-1}H} k_1^{L\sigma_{k-1}}.$$

Doing the same calculation when the current state is L gives us

$$b_1^{H\sigma_{k-1}} + r^{\sigma_{k-1}L} k_1^{H\sigma_{k-1}} = b_1^{L\sigma_{k-1}} + r^{\sigma_{k-1}L} k_1^{L\sigma_{k-1}}.$$

Subtracting the second equation from the first and rearranging then yields

$$\left[r^{\sigma_{k-1}H} - r^{\sigma_{k-1}L}\right] k_1^{H\sigma_{k-1}} = \left[r^{\sigma_{k-1}H} - r^{\sigma_{k-1}L}\right] k_1^{L\sigma_{k-1}}$$

which then implies that either $r^{\sigma_{k-1}H} = r^{\sigma_{k-1}L}$ or $k_1^{H\sigma_{k-1}} = k_1^{L\sigma_{k-1}}$. In either case, we conclude that r or k_1 depends not on σ_k but rather on σ_{k-1}. Iterating this argument

[5]This analysis was first demonstrated by Orrego and Spear (2011).

backwards, we end up at the strongly stationary case, so the argument there then establishes that there cannot be a short memory REE. □

Note that this argument will fail to hold then $k = \infty$. As noted previously, Woodford (1986) showed how to prove existence when the state variable was a full infinite sequence of shocks. Spear and Srivastava (1986) and Duffie et al. (1994) showed how to find Markovian equilibrium in which lagged endogenous state variables served as sufficient statistics for the full infinite history of shocks. Ríos-Rull (1996) simulated production economies with 70 period lived OLG households under the assumption that the wealth distribution was the correct set of lagged endogenous state variables. Citanna and Siconolfi (2010) demonstrated that, in fact, the lagged wealth distribution would be sufficient for this kind of Markovian equilibrium generically. Hence, we work with Rios-Rull's formulation and take the lagged asset holdings of the agents together with the current shock as the set of state variables.

This example makes clear that asset-market incompleteness is not the source of history dependence: even with a sequentially complete asset structure — capital plus a bond spanning both states — the equilibrium remains history dependent, because history dependence arises from the SOLG structure of finite lives and portfolio rebalancing, not from any restriction on the span of tradeable assets.

The finite lives of individual households are particularly important. In representative-agent DSGE models, portfolio rebalancing across implicit generations within the household generates income effects, but the infinite planning horizon allows these effects to be internalized through adjustments to bequests, so that intertemporal allocations remain history independent. Since OLG households leave no bequests, this option is unavailable to them. This is also true of heterogeneous agent models with dynastic household when markets are complete and the Negishi approach to characterizing equilibrium is valid. Genuinely heterogeneous-agent models such as HANK are a different matter: these models are built on the premise that markets are incomplete and idiosyncratic risk is uninsurable, so heterogeneity is *not* fully internalized into a single dynastic Euler equation, and the cross-sectional wealth distribution plays an independent role much as it does in our SOLG environment. Even so, the dynastic households in these models have the bequest mechanism available to them, which leads us to conjecture that the outputs of such model will be smoother than what one sees in the SOLG environment.

This finite-lives/no-bequest distinction is not the only respect in which SOLG models depart from the standard DSGE benchmark, however; the asset structures appropriate to the two frameworks differ as well. In the standard DSGE approach with complete markets, Arrow's theorem on the equivalence of current trading REE and Arrow-Debreu equilibrium implies that the usual Radner equilibrium is equivalent to that obtained by changing the asset structure to one of bonds and self-financing Arrow securities (with firms financing capital accumulations via debt). This equivalence breaks down in the SOLG model. Henriksen and Spear (2012) show, in the context of a three-period pure exchange SOLG model with Lucas tree assets, that one can obtain a Pareto optimal stochastic steady-state equilibrium using the bonds and self-financing Arrow securities framework.[6] This equilibrium is strongly stationary for i.i.d. shocks (and will inherit whatever memory properties are built into the exogenous shock process). They also show that any Pareto optimal allocation would necessarily be strongly stationary, so that the competitive equilibrium is not Pareto optimal, even with sequentially complete asset markets. Again, this is a key difference between SOLG and models based on dynastic households.

[6] The self-financing assumption on the Arrow securities eliminates the income effects associated with portfolio rebalancing.

2.4. Keynesian Search

Parallel with Farmer's construction in the conventional DSGE framework, the timing of production and consumption activities in the model is as follows. First, firms hire seasoned labor in the middle-aged market at the competitive market-clearing wage. Since the labor market for season workers is complete, the middle-aged households are fully employed at this wage. Firms then forecast how much they believe they will need to produce and allocate some measure n_{21} to direct production activities, with the remainder $n_{22} = 1 - n_{21}$ allocated to searching for entry-level workers.

The search technology for recruiting young workers combines the number of young workers searching for jobs with the number of seasoned workers allocated to search and determines the measure of employment matches. The assumption that young agents all search is consistent with Farmer's assumption that workers are all hired anew in every period. It is also consistent with our notion of entry-level labor in the three-period OLG setting, though we adjust this in the multi-period setting to be more realistic. This process then produces job matches according to

$$n_1 = M\left(\Gamma n_{22}, 1\right) \tag{19}$$

The matching function M has the usual properties and, following Farmer, Γ denotes the recruiting productivity of middle-aged agents assigned to recruiting.

The representative firm's static profit maximization problem is

$$\max_{n_{21t}, n_{22t}, k_t} f\left(q_t n_{22t}, n_{21t}, k_t\right) - w_{mt}\left[n_{21t} + n_{22t}\right] - w_{yt} q_t n_{22t} - r_t k_t \tag{20}$$

where q_t is a measure of the recruitment productivity that the individual firm takes as given, but which will be determined in equilibrium from the search process. Specifically, given the matching function and the recruiting productivity parameter Γ, allocating n_{22t} mass of middle-aged labor to search (the firm's search intensity) yields $q_t n_{22t}$ in young labor input for the firm, and the firm takes q_t as given. As in Farmer, we can interpret this parameter as reflecting search costs due to congestion in the labor market for young workers. Under the assumption that markets are competitive, if the production function is Cobb-Douglas,

$$y = B\left(q n_{22}\right)^{\alpha_1} n_{21}^{\alpha_2} k^{\alpha_k} \tag{21}$$

then maximizing with respect to the allocation to n_{22} and assuming that middle-aged agents are paid the same wage, the firm's first-order conditions are (ignoring time subscripts)

$$Bqf_1\left(\bullet\right) - qw_1 - w_2 = 0 \tag{22}$$

$$Bf_2\left(\bullet\right) - w_2 = 0 \tag{23}$$

$$Bf_3\left(\bullet\right) - r = 0 \tag{24}$$

or

$$\frac{\alpha_1}{n_{22}} y = qw_1 + w_2 \tag{25}$$

$$\frac{\alpha_2}{n_{21}} y = w_2 \tag{26}$$

$$\frac{\alpha_k}{k} y = r. \tag{27}$$

Since firms meet demand and labor markets are competitive, output y and the wage variables in these equations are taken as given. This gives us three equations to determine the allocation of middle-aged workers between recruiting and management, plus capital. Focusing on the labor markets for the Cobb-Douglas production function, the first two Euler equations give

$$\frac{\alpha_1 y}{n_{22}} - \frac{\alpha_2 y}{1 - n_{22}} - q w_1 = 0.$$

If we let $x = \frac{1}{n_{22}}$ this reduces to a quadratic of the form

$$\alpha_1 y x^2 - \left[(\alpha_1 + \alpha_2)\, y + q w_1\right] x + q w_1 = 0$$

or

$$x^2 - \left[1 + \frac{\alpha_2}{\alpha_1} + \frac{q w_1}{\alpha_1 y}\right] x + \frac{q w_1}{\alpha_1 y} = 0 \tag{28}$$

In the search equilibrium, we determine q via $q n_{2m} = M\left(\Gamma n_{2m}, 1\right)$ or $q = M\left(\Gamma, \frac{1}{n_{2m}}\right)$ via the degree one homogeneity assumption that is standard for matching functions. Since the second-order condition matrix can be shown to be negative definite, the positive solutions to equation (23) will be local maxima for the profit maximization problem. In a standard benchmark in which wages clear both labor markets, the general equilibrium would determine wages, employment, and output jointly.

The general equilibrium in the model would then determine the market-clearing wages and output. However, with the incomplete entry-level labor market assumption, employment of young agents will be determined once firms know the demand for output, and wages will then be determined once employment levels are known. The determination of overall demand in the model will then depend, as in Farmer's work, on household expectations of future income and wealth. Indeed, in Farmer's DSGE framework, we would choose an expectations function for asset values, which, together with the wages generated on the production side, would completely determine agents' wealth and hence, via standard utility maximization, their demands for output and savings. Given the expectations, firms' employment and production decisions would then be consistent with expectations, so that the resulting equilibrium would be a rational expectations equilibrium.

As we have noted previously, this will not be the case in general in the stochastic OLG framework. The key difference between the two models is that in the SOLG economy the wealth distribution is not just an outcome of equilibrium; it is part of the state on which equilibrium forecasts depend. A candidate belief rule generates household decisions, firm decisions, prices, and hence a new cross-sectional distribution of asset holdings. A rational-expectations equilibrium requires that this induced distribution coincide with the distribution used by agents when forming their forecasts. Thus the SOLG version of Keynesian search replaces Farmer's arbitrary belief function with a fixed-point problem between beliefs, temporary equilibrium allocations, and the law of motion for the wealth distribution

2.5. Keynesian Search in SOLG

In general, working with Farmer's algorithm of first specifying expectations and then determining household consumption and firm production activities will be intractable in the SOLG setting. But, the model will still support Farmer's contention that expectations cause movements in real economic variables in the rational expectations equilibrium by reversing the causality and first determining firm activities, given the demand for output and capital. If we fix capital for the moment (and assume that households save by purchasing equity shares in the firms which entitle them to shares in firms' profits), then determining firm activities boils down to specifying an allocation for n_{22}, which then determines wages and employment. This determines household income, so that solving the model completely now involves finding the competitive equilibrium for a simple pure exchange economy.

In the general stochastic setting, we can drive the model by assuming that $n_1 = \tilde{n}_1$ follows a given stochastic process. The simplest way to do this is to assume that n_1 fluctuates at business cycle frequencies. This then determines at each time t, $n_{2m} = \frac{1}{q}\tilde{n}_1$ determined as above and $n_{1m} = 1 - n_{2m}$, and the output determined from the production function. The wages are then given from the marginal productivity conditions. We can now find the REE for this model by looking at the related pure exchange model in which young agents have endowments equal to $w_1 n_1$ while middle-aged agents have endowments w_2. In our actual simulations, we assume that the productivity parameter q is random, with n_{21} and n_{22} fixed (at calibrated values), so that variations in recruiting productivity drive variations in income.[7] Channeling this variation through production and wages generates the income process for the SOLG product market, which in turn determines the REE expectations that will lead firms to optimally allocate recruiting resources in the way we have posited. We will also show below that when capital is variable, the Keynesian search process leads to a reduced-form model in which household saving is done via the purchase or sale of equity shares in firms, while firms own capital and accumulate it optimally. This model looks like a standard SOLG model with capital.

2.6. The 58-period model

The three-period model isolates the main economic mechanism, but each model period corresponds to roughly twenty years. This aggregation is useful for exposition but too coarse for studying annual business-cycle fluctuations, lifecycle employment transitions, or realistic asset-holding decisions. The 58-period model is therefore the annual-frequency implementation of the same mechanism. It allows entry-level employment, portfolio rebalancing, and retirement saving to evolve at a frequency closer to the data.

We now describe the annual-frequency model used for our quantitative simulations. Time is discrete and agents live for $H = 58$ periods, entering the market at biological age 24 (model age $h = 1$) and dying deterministically at biological age 81 (model age $h = 58$). Agents spend the first 10 periods of life as unseasoned (entry-level) workers subject to unemployment risk; the next 32 periods as seasoned workers with full employment; and the final 16 periods in retirement with zero labor income. Thus, unseasoned working ages are $h \in \{1, \ldots, 10\}$, seasoned working ages are $h \in \{11, \ldots, 42\}$, and retirement ages are $h \in \{43, \ldots, 58\}$.

There is a single perishable consumption good. Households save using a one-period zero-net-supply bond and capital claims. Let φ_t denote the time-t price of the one-period bond, which pays one unit of the consumption good at time $t + 1$. Let

$$R_t \equiv 1 + r_t - \delta$$

denote the gross return to capital received at time t on capital chosen at time $t - 1$.

Throughout the 58-period model, h indexes model age and t indexes calendar time. Thus $c_{h,t}$, $b_{h,t}$, and $k_{h,t}$ denote consumption, bond holdings, and capital holdings chosen by an age-h household at calendar time t. The aggregate state s_t affects households through the age-specific unseasoned employment profile $n_{1,h}(s_t)$, the induced recruiting-productivity objects, and the equilibrium prices φ_t, $w_{1,t}$, $w_{2,t}$, and R_t. We impose entry and terminal asset conditions

$$b_{0,t-1} = k_{0,t-1} = 0, \qquad b_{58,t} = k_{58,t} = 0.$$

Households incur a quadratic cost when adjusting their capital holdings. Specifically, changing capital holdings from $k_{h-1,t-1}$ to $k_{h,t}$ costs

$$\psi_k \left(k_{h,t} - k_{h-1,t-1}\right)^2 .$$

[7] Effectively, the changes in recruiting productivity drive fluctuations in the Beveridge curve.

This term smooths large portfolio adjustments across ages and also improves the numerical stability of the solution. Setting $\psi_k = 0$ recovers the frictionless capital-accumulation problem.

Preferences. Each household has time-separable CRRA preferences

$$u(c) = \begin{cases} \dfrac{c^{1-\gamma}-1}{1-\gamma}, & \gamma \neq 1, \\ \log c, & \gamma = 1, \end{cases} \tag{29}$$

and discounts future utility at factor $\beta \in (0,1)$.

Household problem. Consider a household born at calendar date t. At model age h, the household is alive at calendar date $t+h-1$. The household chooses

$$\{c_{h,t+h-1},\, b_{h,t+h-1},\, k_{h,t+h-1}\}_{h=1}^{58}$$

to maximize

$$\max_{\{c_{h,t+h-1},\, b_{h,t+h-1},\, k_{h,t+h-1}\}_{h=1}^{58}} \mathbb{E}_t \left[\sum_{h=1}^{58} \beta^{h-1} u(c_{h,t+h-1}) \right], \tag{30}$$

subject to the sequence of age-specific budget constraints below.

For unseasoned workers, $h = 1, \ldots, L_{\text{young}}$, with $L_{\text{young}} = 10$,

$$\begin{aligned} c_{h,t+h-1} = {} & w_{1,t+h-1} n_{1,h}(s_{t+h-1}) - \varphi_{t+h-1} b_{h,t+h-1} + b_{h-1,t+h-2} \\ & k_{h,t+h-1} + R_{t+h-1} k_{h-1,t+h-2} - -\psi_k \big(k_{h,t+h-1} - k_{h-1,t+h-2}\big)^2. \end{aligned} \tag{31}$$

For seasoned workers, $h = L_{\text{young}}+1, \ldots, L_{\text{young}} + L_{\text{middle}}$, with $L_{\text{middle}} = 32$,

$$\begin{aligned} c_{h,t+h-1} = {} & w_{2,t+h-1} - \varphi_{t+h-1} b_{h,t+h-1} + b_{h-1,t+h-2} \\ & - k_{h,t+h-1} + R_{t+h-1} k_{h-1,t+h-2} - \psi_k \big(k_{h,t+h-1} - k_{h-1,t+h-2}\big)^2. \end{aligned} \tag{32}$$

For retired households, $h = L_{\text{young}} + L_{\text{middle}} + 1, \ldots, 58$,

$$\begin{aligned} c_{h,t+h-1} = {} & - \varphi_{t+h-1} b_{h,t+h-1} + b_{h-1,t+h-2} \\ & - k_{h,t+h-1} + R_{t+h-1} k_{h-1,t+h-2} - \psi_k \big(k_{h,t+h-1} - k_{h-1,t+h-2}\big)^2. \end{aligned} \tag{33}$$

The entry and terminal asset conditions are

$$b_{0,t-1} = k_{0,t-1} = 0, \qquad b_{58,t+57} = k_{58,t+57} = 0. \tag{34}$$

In (31)–(33), $w_{1,t+h-1}$ is the unseasoned wage, $w_{2,t+h-1}$ is the seasoned wage, and $n_{1,h}(s_{t+h-1}) \in [0,1]$ is the age-specific unseasoned employment/hours profile. Seasoned workers are fully employed and supply one unit inelastically.

Borrowing/feasibility constraints. To ensure feasibility and to match standard asset-pricing implications in lifecycle models[8], we impose in the simulations

$$c_{h,t+h-1} \geq 0, \qquad k_{h,t+h-1} \geq 0, \qquad b_{h,t+h-1} \geq \underline{b}, \qquad h = 1, \ldots, 58. \tag{35}$$

where $\underline{b}$ is a borrowing limit.

[8] See Hasanhodzic (2014) or Hasanhodzic and Kotlikoff (2025) for details.

Euler equations. With the inequality constraints $b_t^\tau \geq \underline{b}$ and $k_t^\tau \geq 0$, the necessary conditions can be written as complementarity (KKT) conditions. For $\tau = t, \dots, t+56$,

$$0 \;\leq\; u'(c_t^\tau)\;\varphi_\tau - \beta\,\mathbb{E}_\tau\big[u'\big(c_t^{\tau+1}\big)\big] \;\perp\; b_t^\tau - \underline{b} \;\geq\; 0, \tag{36}$$

$$0 \;\leq\; \Big(1 + 2\psi_k\big(k_t^\tau - k_t^{\tau-1}\big)\Big)\,u'(c_t^\tau) - \beta\,\mathbb{E}_\tau\Big[u'\big(c_t^{\tau+1}\big)\,\Big(1 + r_{t+1} - \delta + 2\psi_k\big(k_t^{\tau+1} - k_t^\tau\big)\Big)\Big] \;\perp\; k_t^\tau \;\geq\; 0, \tag{37}$$

where $\mathbb{E}_\tau[\cdot]$ conditions on information at calendar time τ (including the current business-cycle regime). For $\psi_k = 0$, (42) reduces to the standard interior Euler equations.

In the numerical solution we enforce complementarity using the Fischer–Burmeister function

$$\psi_{\mathrm{FB}}(x, y) \equiv \sqrt{x^2 + y^2} - x - y, \tag{38}$$

so that (41)–(42) are equivalently imposed as the smooth equalities

$$\psi_{\mathrm{FB}}\Big(u'(c_t^\tau)\;\varphi_\tau - \beta\,\mathbb{E}_\tau\big[u'\big(c_t^{\tau+1}\big)\big]\,,\; b_t^\tau - \underline{b}\Big) = 0, \tag{39}$$

$$\psi_{\mathrm{FB}}\Big(\Big(1 + 2\psi_k\big(k_t^\tau - k_t^{\tau-1}\big)\Big)\,u'(c_t^\tau) - \beta\,\mathbb{E}_\tau\Big[u'\big(c_t^{\tau+1}\big)\,\Big(1 + r_{t+1} - \delta + 2\psi_k\big(k_t^{\tau+1} - k_t^\tau\big)\Big)\Big]\,,\; k_t^\tau\Big) = 0. \tag{40}$$

Euler equations. With the inequality constraints $b_{h,t+h-1} \geq \underline{b}$ and $k_{h,t+h-1} \geq 0$, the necessary conditions can be written as complementarity conditions. For $h = 1, \dots, 57$,

$$0 \;\leq\; u'(c_{h,t+h-1})\,\varphi_{t+h-1} - \beta\,\mathbb{E}_{t+h-1}[u'(c_{h+1,t+h})] \;\perp\; b_{h,t+h-1} - \underline{b} \;\geq\; 0, \tag{41}$$

and

$$\begin{aligned} 0 \leq{}& \Big(1 + 2\psi_k\big(k_{h,t+h-1} - k_{h-1,t+h-2}\big)\Big)u'(c_{h,t+h-1}) \\ &- \beta\,\mathbb{E}_{t+h-1}\Big[u'(c_{h+1,t+h})\,\Big(R_{t+h} + 2\psi_k\big(k_{h+1,t+h} - k_{h,t+h-1}\big)\Big)\Big] \\ &\perp\; k_{h,t+h-1} \geq 0. \end{aligned} \tag{42}$$

where $\mathbb{E}_{t+h-1}[\cdot]$ conditions on information at calendar time $t+h-1$, including the current business-cycle regime. For $\psi_k = 0$, (42) reduces to the standard interior Euler equations.

In the numerical solution we enforce complementarity using the Fischer–Burmeister function

$$\psi_{\mathrm{FB}}(x, y) \equiv \sqrt{x^2 + y^2} - x - y, \tag{43}$$

so that (41)–(42) are equivalently imposed as the smooth equalities

$$\psi_{\mathrm{FB}}\Big(u'(c_{h,t+h-1})\,\varphi_{t+h-1} - \beta\,\mathbb{E}_{t+h-1}[u'(c_{h+1,t+h})]\,,\; b_{h,t+h-1} - \underline{b}\Big) = 0, \qquad h = 1, \dots, 57, \tag{44}$$

$$\begin{aligned} &\psi_{\mathrm{FB}}\bigg(\Big(1 + 2\psi_k\big(k_{h,t+h-1} - k_{h-1,t+h-2}\big)\Big)u'(c_{h,t+h-1}) - \\ &\beta\,\mathbb{E}_{t+h-1}\Big[u'(c_{h+1,t+h})\,\Big(R_{t+h} + 2\psi_k\big(k_{h+1,t+h} - k_{h,t+h-1}\big)\Big)\Big]\,, \\ &k_{h,t+h-1}\bigg) = 0, \qquad h = 1, \dots, 57. \end{aligned} \tag{45}$$

For the recursive equilibrium and numerical implementation, it is useful to relabel variables by the calendar date at which they are observed. Thus $c_{h,t}$, $b_{h,t}$, and $k_{h,t}$ denote the consumption and end-of-period asset choices of the age-h household alive at calendar date t.

Production. The representative firm produces using unseasoned labor, seasoned labor, and capital, as in the three-period model. In the 58-period implementation, the unseasoned block consists of the first $L_{\text{young}} = 10$ working ages. For each aggregate state s_t, we impose an age-specific unseasoned employment profile

$$\mathbf{n}_{1,t} = \left(n_{1,1,t}, \ldots, n_{1,L_{\text{young}},t}\right), \qquad n_{1,h,t} \equiv n_{1,h}(s_t), \quad h = 1, \ldots, L_{\text{young}}.$$

Here the first subscript in $n_{1,h,t}$ denotes unseasoned labor, h denotes model age, and t denotes calendar time. The scalar unseasoned labor input entering aggregate production is the average employment of these young cohorts,

$$\bar{n}_{1,t} = \frac{1}{L_{\text{young}}} \sum_{h=1}^{L_{\text{young}}} n_{1,h,t}. \tag{46}$$

Seasoned labor is normalized to one in per capita units. Firms allocate this scalar supply between production and recruiting:

$$n_{21,t} + n_{22,t} = 1. \tag{47}$$

Here $n_{21,t}$ denotes seasoned labor allocated to production and $n_{22,t}$ denotes seasoned labor allocated to recruiting.

Thus the age-specific vector $\mathbf{n}_{1,t}$ is not an additional firm choice. Given $n_{22,t}$, define the state-dependent average recruiting productivity by

$$\bar{q}_t = \frac{\bar{n}_{1,t}}{n_{22,t}}, \qquad \bar{n}_{1,t} = \bar{q}_t n_{22,t}.$$

With Cobb–Douglas technology, the production function used in the simulations is

$$y_t = B\left(\bar{n}_{1,t}\right)^{\alpha_1}\left(n_{21,t}\right)^{\alpha_2}\left(\frac{K_{t-1}}{L_{\text{young}} + L_{\text{middle}}}\right)^{\alpha_k}, \qquad \alpha_1 + \alpha_2 + \alpha_k = 1. \tag{48}$$

Equivalently, since $\bar{n}_{1,t} = \bar{q}_t n_{22,t}$,

$$y_t = B\left(\bar{q}_t n_{22,t}\right)^{\alpha_1}\left(n_{21,t}\right)^{\alpha_2}\left(\frac{K_{t-1}}{L_{\text{young}} + L_{\text{middle}}}\right)^{\alpha_k}.$$

Under perfect competition, factor prices satisfy marginal productivity pricing. In particular,

$$w_{2,t} = \frac{\alpha_2 y_t}{n_{21,t}}, \qquad w_{1,t} = \frac{\alpha_1 y_t / n_{22,t} - w_{2,t}}{\bar{q}_t}.$$

The gross return to capital satisfies

$$R_t = 1 + r_t - \delta, \qquad r_t = \frac{\alpha_k y_t}{k_{t-1}}, \tag{49}$$

where

$$k_{t-1} = \frac{K_{t-1}}{L_{\text{young}} + L_{\text{middle}}}.$$

Aggregation and market clearing. Aggregate consumption and end-of-period asset holdings are

$$C_t \equiv \sum_{h=1}^{58} c_{h,t}, \qquad B_t \equiv \sum_{h=1}^{58} b_{h,t}, \qquad K_t \equiv \sum_{h=1}^{58} k_{h,t}.$$

Bond market clearing requires

$$B_t = 0. \tag{50}$$

Aggregate capital used in production at time $t+1$ is K_t, so production at time t uses the inherited aggregate capital stock K_{t-1}.

Labor-market aggregation is defined in the production block. In particular, the scalar unseasoned labor input is given by (46), and the normalized seasoned-labor allocation satisfies (47).

3. Simulations

3.1. Calibration

This section calibrates the annual-frequency 58-period economy described in Section 2.6. Agents enter at age 24 ($h = 1$) and live for $H = 58$ periods.

Preferences. We set the annual discount factor to $\beta = 0.97$ and risk aversion to $\gamma = 2$.

Technology. We assume that production exhibits constant returns to scale in the two types of labor and capital. Based on this, we use the standard factor shares approach to calibrate the exponents in the Cobb–Douglas production function

$$y_t = B\,\bar{n}_{1,t}^{\alpha_1} n_{21,t}^{\alpha_2} k_{t-1}^{\alpha_k},$$

where $\bar{n}_{1,t}$ is the average unseasoned labor input and $n_{21,t}$ is seasoned labor allocated to production. For low-skilled workers, estimates of their shares are around 22% (see the Economic Policy Institute (n.d.) online share calculator), so we take $\alpha_1 = 0.22$. With overall labor share at 66%, this implies that $\alpha_2 = 0.44$ with the capital share at the usual $\alpha_k = 0.33$. Capital depreciates at $\delta = 0.05$ annually. We set the quadratic capital-adjustment-cost parameter to $\psi_k = 0.2$. This modest adjustment cost smooths large age-to-age changes in capital holdings and improves numerical stability; setting $\psi_k = 0$ recovers the frictionless benchmark.

Aggregate shock process. Business-cycle states follow a four-state Markov chain with two recession states and two expansion states. The annual transition matrix, based on the Chauvet-Guo regime-switching calibration, is reported in Table 1.

	S_1	S_2	S_3	S_4
S_1	0.445	0.445	0.030	0.030
S_2	0.445	0.445	0.030	0.030
S_3	0.055	0.055	0.470	0.470
S_4	0.055	0.055	0.470	0.470

Table 1: Annual transition matrix for the four-state business-cycle process. States S_1 and S_2 are recession states, while S_3 and S_4 are expansion states.

States S_1 and S_2 are recession states, and states S_3 and S_4 are expansion states. With this transition matrix, the economy is in a recession state one third of the time and in an expansion state two thirds of the time.

Keynesian search. Given the search procedure for determining employment of the young workers, we need to calculate how much the intensity of the firm's recruitment must fluctuate to generate the standard business cycle fluctuations of 3%. The simplest way to model this is to assume that firm recruiting productivity is subject to shocks, i.e., to fluctuations in the average recruiting-productivity parameter $\bar{q}$. We note, however, that in actual firms, HR hiring productivity is likely endogenous, depending on the firm's expectations of its labor needs and output demand. Absent a detailed theory of how these expectations are formed, we follow the widely applied approach of simply assuming that these are stochastic.

This recruiting-productivity interpretation connects the calibration to recent work on Beveridge-curve shifts and labor-market tightness. In that literature, changes in unemployment and vacancies reflect not only movements along a stable matching function, but also shifts in matching efficiency, worker search behavior, sectoral reallocation, and labor-force participation. Our $\bar{q}(s_t)$ process is a reduced-form way to introduce these forces into a lifecycle general-equilibrium environment.

We should emphasize that our reduced form approach to driving business cycle fluctuations is at odds with other interpretations for the shifting of the Beveridge curve; see Barlevy et al. (2024) for details. There is an interesting possible variation on our model that one might use to examine the labor market dynamics more carefully. This would entail fixing the wages of unseasoned workers and then letting the REE demand determine employment levels for these workers. One could then empirically examine whether a Beveridge curve relationship exists contingent on the state of the (exogenously imposed) business cycle. We leave this to future research, however.

To discipline recruiting *intensity*, we use HR/employee ratios from https://www.sesamehr.com/blog/hr-to-employee-ratio/ (figures are per 100 employees):

- Fewer than 100 employees: 2.70
- 100 to 249 employees: 1.26
- 250 to 499 employees: 1.07
- 500 to 999 employees: 0.82
- 1,000 to 2,499 employees: 0.79
- 2,500 to 7,499 employees: 0.53
- 7,500 or more employees: 0.42

Since we are treating the firm as representative of the full economy, we use 0.042–0.07 as the HR/employee ratio, so we take average $n_{22} = 0.05$. Given this n_{22} and the calibrated unseasoned employment profile, the implied state-dependent average recruiting productivity is

$$\bar{q}(s) = \frac{\bar{n}_1(s)}{n_{22}}, \qquad \bar{n}_1(s) = \frac{1}{L_{\text{young}}} \sum_{h=1}^{L_{\text{young}}} n_{1,h}(s).$$

Age-specific ratios $n_{1,h}(s)/n_{22}$ can be computed mechanically, but these should be interpreted only as a way of reporting the calibrated age profile, not as separate firm-level recruiting choices.

Our implementation uses a Keynesian-search reduced form in which firms allocate a scalar measure $n_{22,t}$ of seasoned workers to recruiting. In the simulations, we hold this allocation fixed at $n_{22,t} = n_{22} = 0.05$, so that $n_{21,t} = n_{21} = 1 - n_{22} = 0.95$. In the 58-period simulation, however, unseasoned workers occupy the first $L_{\text{young}} = 10$ working ages. We therefore calibrate a state-contingent age profile

$$\mathbf{n}_1(s) = \left(n_{1,1}(s), \ldots, n_{1,L_{\text{young}}}(s)\right), \qquad s \in \{S_1, S_2, S_3, S_4\}.$$

At calendar time t, this profile is evaluated at the realized aggregate state s_t, so that

$$n_{1,h,t} \equiv n_{1,h}(s_t), \qquad \bar{n}_{1,t} \equiv \bar{n}_1(s_t).$$

The scalar unseasoned labor input entering production is

$$\bar{n}_1(s) = \frac{1}{L_{\text{young}}} \sum_{h=1}^{L_{\text{young}}} n_{1,h}(s).$$

Given the fixed recruiting allocation n_{22}, the implied state-dependent average recruiting productivity is

$$\bar{q}(s) = \frac{\bar{n}_1(s)}{n_{22}}.$$

The age-specific objects $n_{1,h}(s)$ are therefore not separate firm-level recruiting choices. They are calibrated age-specific employment/hours profiles used to distribute the scalar unseasoned-labor outcome across the first ten working ages.

To discipline the age profile of unseasoned employment/hours, we calibrate the first 10 working ages using the lifecycle evidence in Ríos-Rull (1996). Figure 1 plots the exogenously imposed age–labor endowment schedule used in the simulations. For unseasoned ages $h = 1, \dots, 10$ (biological ages 24–33), labor endowments are state-contingent and given by $\{n_{1,h}(s)\}_{h=1}^{10}$, constructed from the Ríos-Rull baseline profile and the regime-dependent scaling described below. For seasoned working ages $h = 11, \dots, 42$ (biological ages 34–65), we normalize the labor endowment to one, and for retirement ages $h \geq 43$ (biological ages ≥ 66) we set labor endowment to zero. Differences across regimes therefore operate entirely through the unseasoned block.

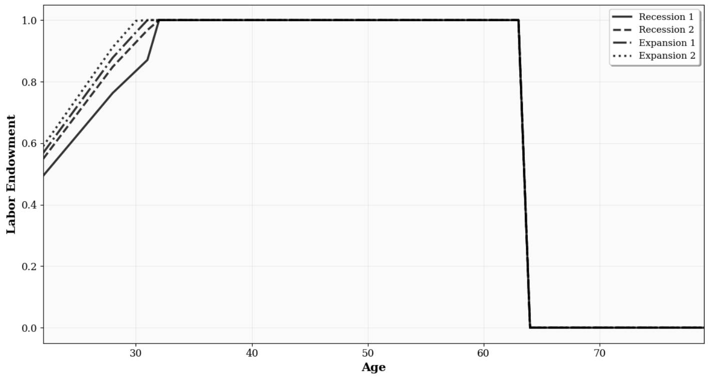


Figure 1: State-contingent labor endowment schedule by age. For unseasoned ages 24–33 ($h = 1, \dots, 10$), endowments are regime-dependent and equal to $n_{1,h}(s)$. For seasoned working ages 34–65 ($h = 11, \dots, 42$), endowments are normalized to $n_{2,h} = 1$, and for retirement ages ≥ 66 ($h \geq 43$) endowments are $n_{r,h} = 0$.

For the calibration of the recruiting-productivity wedge, start from the production representation implied by the search block,

$$y = B\, \bar{n}_1^{\alpha_1}\, n_{21}^{\alpha_2}\, k^{\alpha_k} = B\, (\bar{q}\, n_{22})^{\alpha_1}\, n_{21}^{\alpha_2}\, k^{\alpha_k}, \qquad \text{with } n_{21} = 1 - n_{22}.$$

Taking logs gives

$$\ln y = \ln B + \alpha_1 \ln \bar{q} + \alpha_1 \ln n_{22} + \alpha_2 \ln n_{21} + \alpha_k \ln k. \tag{51}$$

Holding n_{22} and n_{21} fixed in the calibration step implies

$$\frac{dy}{y} = \alpha_1 \frac{d\bar{q}}{\bar{q}} + \alpha_k \frac{dk}{k}. \tag{52}$$

With $\frac{\alpha_k y}{k} = r$ from the first-order condition for capital, we get

$$dk = \frac{\alpha_k}{r} dy - \frac{\alpha_k y}{r^2} dr,$$

so that

$$\frac{dk}{k} = \frac{\frac{\alpha_k}{r} dy}{\frac{\alpha_k y}{r}} - \frac{\frac{\alpha_k y}{r^2} dr}{\frac{\alpha_k y}{r}} = \frac{dy}{y} - \frac{dr}{r}.$$

Putting this together with the previous expression, we have

$$\frac{d\bar{q}}{\bar{q}} = \frac{1}{\alpha_1} \left[(1 - \alpha_k) \frac{dy}{y} + \alpha_k \frac{dr}{r} \right]. \tag{53}$$

Since r is determined endogenously in the model, we can handle calibration by picking a "ballpark" value (based on variability of real long-run interest rates) and then use the simulation based on this calibration to determine a new number for the bond price variance and iterate on this.

From this, for output fluctuations totaling 3%, and taking the volatility of asset prices to be 1% (based on recent data from DataTrek, https://datatrekresearch.com/long-term-look-at-us-equity-volatility/?v=7516fd43adaa), the implied proportional recruiting-productivity shift is

$$\eta \equiv \frac{d\bar{q}}{\bar{q}} = \frac{1}{0.22}\left[(0.66)\,(0.03) + (0.33)\,(0.01)\right].$$

We apply this proportional shift to the baseline unseasoned age profile. For each unseasoned age $h = 1, \ldots, 10$, the regime-dependent employment profile is generated by

$$n_{1,h}(s) = \bar{n}_{1,h}\left(1 + \sigma_s \frac{\eta}{2} + \omega_s\right),$$

where $\sigma_s = -1$ for the recession states S_1, S_2 and $\sigma_s = 1$ for the expansion states S_3, S_4. The within-phase adjustment is $\omega_s = \pm 0.05$ for the two recession states and $\omega_s = \pm 0.02$ for the two expansion states, with the negative sign for S_1, S_3 and the positive sign for S_2, S_4. Values are capped at one when necessary.

We interpret this as a symmetric phase shift around the baseline profile. Using the Chauvet and Guo (2003) estimates of conditional standard deviations in expansions and recessions of 2% and 5%, respectively, we generate four regime-dependent unseasoned employment schedules. For each unseasoned age $h = 1, \ldots, 10$, the regime-dependent employment profile is

$$n_{1,h}(s) = \bar{n}_{1,h}\left(1 + \sigma_s \frac{\eta}{2} + \omega_s\right),$$

where $\sigma_s = -1$ in the recession states S_1, S_2 and $\sigma_s = 1$ in the expansion states S_3, S_4. The within-phase adjustment is

$$\omega_s = \begin{cases} -0.05, & s = S_1, \\ 0.05, & s = S_2, \\ -0.02, & s = S_3, \\ 0.02, & s = S_4. \end{cases}$$

Values are capped at one when necessary. For seasoned working ages $h = 11, \ldots, 42$, labor endowment is normalized to one, while for retirement ages $h = 43, \ldots, 58$, it is set to zero. This construction preserves the Ríos-Rull lifecycle profile while allowing regime switching to generate aggregate fluctuations through shocks to recruiting productivity.

Wages. Since firms produce to meet expected demand, we can take output y_t as given and use the parameterizations above to compute wages. The unseasoned wage bill follows from the firm's first-order conditions for recruiting and seasoned labor. These conditions imply

$$\frac{\alpha_1 y_t}{n_{22,t}} = \bar{q}_t w_{1,t} + w_{2,t}, \qquad w_{2,t} = \frac{\alpha_2 y_t}{n_{21,t}}.$$

Multiplying the first condition by $n_{22,t}$ and using $\bar{n}_{1,t} = \bar{q}_t n_{22,t}$ gives

$$\alpha_1 y_t = w_{1,t}\bar{n}_{1,t} + w_{2,t} n_{22,t}.$$

Substituting the seasoned wage therefore yields

$$w_{1,t}\bar{n}_{1,t} = \left[\alpha_1 - \frac{\alpha_2 n_{22,t}}{n_{21,t}}\right] y_t.$$

The subtraction reflects the opportunity cost of diverting seasoned workers from production to recruiting: part of the output share associated with the unseasoned-labor block is used to compensate the seasoned workers engaged in recruiting. Because normalized seasoned labor satisfies $n_{21,t} + n_{22,t} = 1$, its total wage bill is

$$w_{2,t}(n_{21,t} + n_{22,t}) = w_{2,t} = \frac{\alpha_2 y_t}{n_{21,t}}.$$

Thus wages and wage bills are determined by the calibrated factor shares, the recruiting allocation, the unseasoned employment profile, and equilibrium output. The calibration numbers are summarized in Table 2.

If we were following Farmer's qualitative elaboration of the Keynesian search model and omitting capital, this calibration is all we would need, since the resulting calibrated model reduces to a simple pure-exchange economy. When we add capital, however, wages and returns to capital (and via the interactions in the asset markets, returns to bonds) will all be determined endogenously based only on the exogenously imposed shocks to recruiting productivity and the calibrated business cycle transition probabilities.

Calibration Summary	
(1) Demographics	
Biological age at $h = 1$	24 years
Model age at retirement ($h = 43$)	66 years
Maximum model age ($h = 58$)	81 years
(2) Preferences and Technology Parameters	
β (discount factor, annual)	0.97
γ (risk aversion)	2
δ (capital depreciation, annual)	5%
Capital adjustment cost (ψ_k)	0.2
α_1 (young/low-skill labor share)	0.22
α_2 (seasoned labor share)	0.44
α_k (capital share)	0.33
(3) Labor Market and Recruiting	
n_{22} (HR/employee ratio; recruiting intensity)	0.05
Baseline unseasoned profile $\{\bar{n}_{1,h}\}_{h=1}^{10}$	see Fig. 1
	$\bar{n}_1(s) = L_{\text{young}}^{-1} \sum_{h=1}^{L_{\text{young}}} n_{1,h}(s)$
Regime-dependent recruiting productivity	$\bar{q}(s) = \bar{n}_1(s)/n_{22}$
(4) Macro Shocks and Target Volatilities	
Business-cycle states	S_1, S_2 recessions; S_3, S_4 expansions
Transition matrix	see Table 1
Target output fluctuation	3%
Target asset price volatility	1%

Table 2: Summary of calibrated parameters, recruiting calibration, and macro shock targets for the 58-period economy.

3.2. Solution method

Our neural-network solution method follows the deep-learning-based equilibrium-solver approach of Azinovic and Zemlicka (2024), adapted to the present 58-period SOLG setting. Solving the 58-period model requires computing a recursive rational-expectations equilibrium in which agents' optimal decisions are consistent with the endogenously generated wealth distribution. Let the endogenous state be the cross-section of lagged asset holdings across cohorts,

$$\xi_{t-1} \equiv \Big(\{b_{h,t-1}\}_{h=1}^{58},\ \{k_{h,t-1}\}_{h=1}^{58} \Big), \tag{54}$$

together with the current regime $s_t \in \{0, 1, 2, 3\}$. Define the full state as $x_t \equiv (\xi_{t-1}, s_t) \in \mathbb{R}^{116} \times \{0, 1, 2, 3\}$ (117 objects, counting the regime indicator). A recursive equilibrium consists of policy functions for next-period asset choices

$$(b_{h,t}, k_{h,t}) = \mathcal{G}_h(x_t), \qquad h = 1, \ldots, 58, \tag{55}$$

and a bond-price function

$$\varphi_t = \Phi(x_t), \tag{56}$$

such that: (i) given $(\Phi, \{\mathcal{G}_h\}_h)$, households satisfy optimality (Euler equations) and feasibility/borrowing constraints; and (ii) markets clear, including the zero-net-supply bond condition (50). The rational-expectations fixed point is implicit in the requirement that the laws of motion for ξ_t generated by these policies coincide with the beliefs embedded in the Euler expectations operator.

Neural network approximation. We approximate the equilibrium objects $\{\mathcal{G}_h\}_h$ and Φ with neural networks parameterized by weights θ. Concretely,

$$(b_{h,t}, k_{h,t}) = \mathcal{G}_h(x_t; \theta), \qquad \varphi_t = \Phi(x_t; \theta), \tag{57}$$

where inputs include the regime indicator s_t and the lagged wealth distribution ξ_{t-1}, and outputs are age-specific asset choices and the equilibrium bond price.

Architecture-enforced constraints and market clearing. To improve stability and ensure economically admissible choices, we implement key constraints directly in the network outputs. First, we impose nonnegativity of capital holdings by a monotone transform. Second, bond market clearing is enforced by construction: the network produces provisional bond choices and we mean-recenter them so that $\sum_{h=1}^{58} b_{h,t} = 0$ holds exactly (up to numerical precision) at every t. Borrowing limits are enforced through the complementarity (Fischer–Burmeister) residual using the slack $b_{h,t} - \underline{b}$ evaluated on the post-recentered bond choices. We also pin asset holdings of entering and exiting cohorts to zero where appropriate, consistent with the model timing. These architectural restrictions substantially reduce the burden placed on penalty terms and prevent the solver from drifting into infeasible regions.

Expectations and Euler residuals. Euler equations involve conditional expectations over next period's regime. Because s_t follows a finite-state Markov chain with transition matrix Π, we compute expectations exactly by summation:

$$\mathbb{E}_t[\cdot] \equiv \sum_{s' \in \{0,1,2,3\}} \Pi(s'|s_t)\, (\cdot)\big|_{s_{t+1}=s'}.$$

Operationally, given a current state x_t, the network produces $(\varphi_t, \{b_{h,t}, k_{h,t}\}_h)$; we compute prices and allocations implied by the firm's static problem and household budgets; and then evaluate next-period objects under each possible $s_{t+1} = s'$ to form the Euler expectations. Let $\varepsilon^b_{h,t}$ and $\varepsilon^k_{h,t}$ denote the resulting bond and capital Euler residuals implied by (41) and (42). Let $\varepsilon^c_{h,t}$ denote feasibility violations (e.g., $\varepsilon^c_{h,t} = \min\{0, c_{h,t}\}$). We also form an Euler-implied bond price and define the bond-price residual $\varepsilon^\varphi_t \equiv \varphi_t - \varphi^{\text{impl}}_t$.

Loss function and training objective. We train θ to minimize a loss that enforces Euler optimality and feasibility. Since market clearing for bonds is enforced in the architecture, the bond-clearing residual is identically zero up to numerical precision; we therefore do not rely on market-clearing penalties for identification, but can optionally include a small diagnostic penalty if desired. The baseline objective is

$$\min_{\theta} \mathcal{L}(\theta) = \lambda_E \sum_t \sum_{h=1}^{58} \left((\varepsilon^b_{h,t})^2 + (\varepsilon^k_{h,t})^2 \right) + \lambda_C \sum_t \sum_{h=1}^{58} (\varepsilon^c_{h,t})^2 + \lambda_\varphi \sum_t (\varepsilon^\varphi_t)^2, \tag{58}$$

for penalty weights $(\lambda_E, \lambda_C, \lambda_\varphi) > 0$. In implementation, we evaluate (58) on mini-batches of simulated states and use automatic differentiation to compute $\nabla_\theta \mathcal{L}(\theta)$.

Simulation loop, training data, and the RE fixed point. Given an initial distribution ξ_0 and an initial regime s_0, we generate a long simulated panel by iterating:

$$(\xi_{t-1}, s_t) \xrightarrow[\text{NN}]{(\Phi,\mathcal{G})} \left(\varphi_t, \{b_{h,t}, k_{h,t}\}_h\right) \xrightarrow[\text{prices, budgets, aggregation}]{} \xi_t,$$

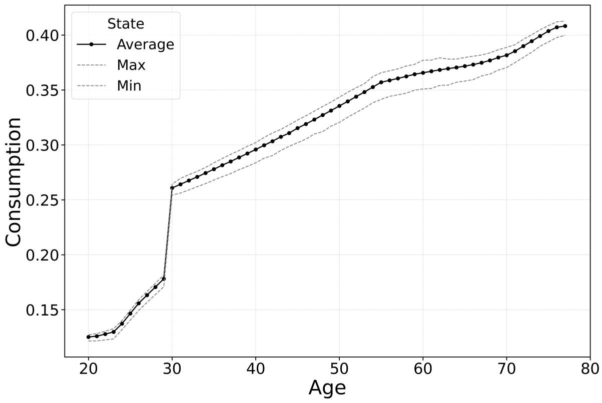
with regimes evolving according to the calibrated Markov transition matrix. The states $\{x_t\}$ produced by this simulation constitute the training distribution: after a burn-in period, they are draws from the economy's long-run (ergodic) distribution under the current candidate policy. We update θ using stochastic gradient methods (Adam) to reduce (58), periodically refreshing the simulated state panel as the policy improves. Convergence is assessed by the magnitude of Euler residuals, bond-price residuals, and feasibility violations along long simulated paths. At convergence, the policy functions used to make decisions generate (through equilibrium pricing and aggregation) the same law of motion for ξ_t that is assumed inside the Euler expectations operator, thus implementing the required rational-expectations fixed point.

Implementation details. All architectural choices (depth/width, activation functions, normalization of inputs/outputs), optimizer settings, batching strategy, staged training/homotopy (if used), and validation checks are reported in the computation appendix. Here we emphasize that the NN procedure is disciplined directly by equilibrium conditions (Euler equations, feasibility, and market clearing), and is therefore a structural solver for the recursive equilibrium of the 58-period economy rather than a reduced-form forecasting exercise.

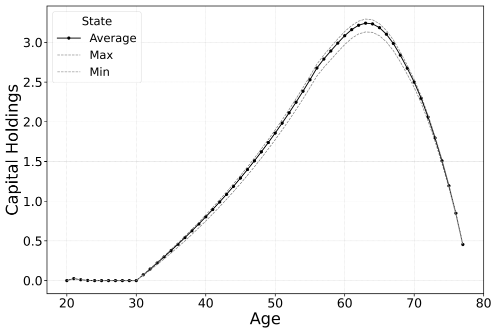
3.3. Quantitative results

The three-period model isolates the main economic mechanism, but each model period corresponds to roughly twenty years. This aggregation is useful for exposition but too coarse for studying annual business-cycle fluctuations, lifecycle employment transitions, or realistic asset-holding decisions. The 58-period model is therefore the annual-frequency implementation of the same mechanism. The quantitative exercise asks whether the fixed-point discipline identified in the three-period model remains operative in a realistically calibrated lifecycle economy. The simulation exercise has three purposes: to compute the recursive rational-expectations fixed point, to recover the implied expectations function, and to examine whether the resulting lifecycle and asset-market dynamics are empirically plausible.

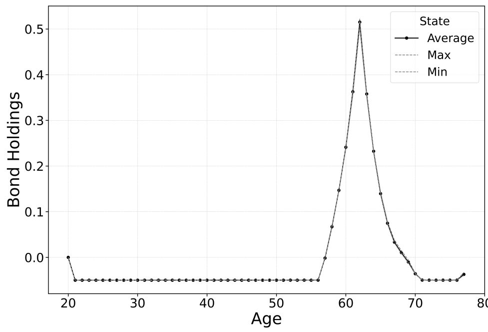
We now summarize key simulation results graphically, with detailed numerical tables provided separately in the online data appendix. To start, Figure 2 presents the lifecycle profiles for average consumption, capital holdings, and bond holdings generated from our model simulations. The consumption profile (Figure 2a) shows monotonically increasing consumption over the lifecycle. While this is at odds with frequent claims that the profile should be hump shaped, we note that much of this claim is driven by data that excludes expenses such as housing. International data is also not uniformly supportive of the hump-shaped profile claim. We note that in these simulations, young agents are credit constrained. This constraint is needed in order to generate realistic equity premiums – see Hasanhodzic (2014) for details. We impose a hard constraints of 0.05 units of consumption (the numeraire good), which amounts to a credit constraint of around 40% of consumption spending. This is well within observed credit limits in the U.S. economy.



(a) Consumption.



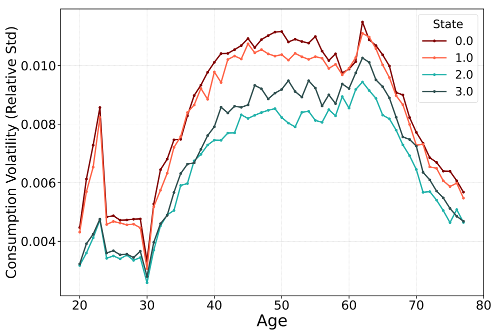
(b) Capital Holdings.



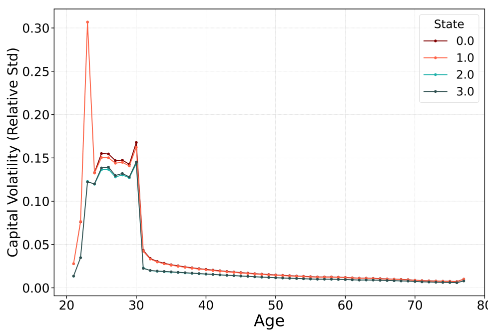
(c) Bond Holdings.

Figure 2: Average Values by Age.

The average capital holdings profile (Figure 2b) matches the empirical data (see, for example, Jappelli and Pistaferri (2017)). The bond holdings profile (Figure 2c) is also typical of what one finds in these kinds of SOLG models (see Hasanhodzic (2014)). It also illustrates the optimality of shifting out of riskier equity assets into less risky but lower yielding bonds as agents age.



(a) Consumption.

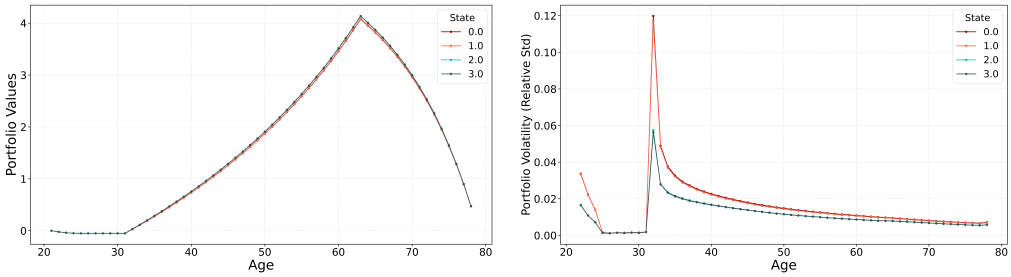


(b) Capital.

Figure 3: Consumption and Capital Volatility by State.

Figure 3a shows the volatility of consumption by age for the four states of the model (where, we recall that states 0 and 1 are recession states, and states 2 and 3 are expansions). The hump-shaped volatility profile is also consistent with the data (see, for example, Storesletten et al. (2007)). For comparison, we also show the volatility profile for capital in Figure 3b. Capital holdings are significantly more volatile than consumption, as expected, and tend to be more volatile early in life.

Since there are a number of articles in the finance literature that examine the returns on portfolios in the phases of the business cycle, we also show, in Figure 4a, the portfolio values for agents by age and state, where this value is defined as$(1 + r)k_{t-1} + b_{t-1}$.



(a) Portfolio Values.

(b) Portfolio Volatility.

Figure 4: Portfolio Values and Volatility by Age and State.

What is striking here is the fact that the optimal portfolio holdings hardly differ over the business cycle. The riskiness of these portfolio holdings does vary over the cycle, as seen in Figure 4b.

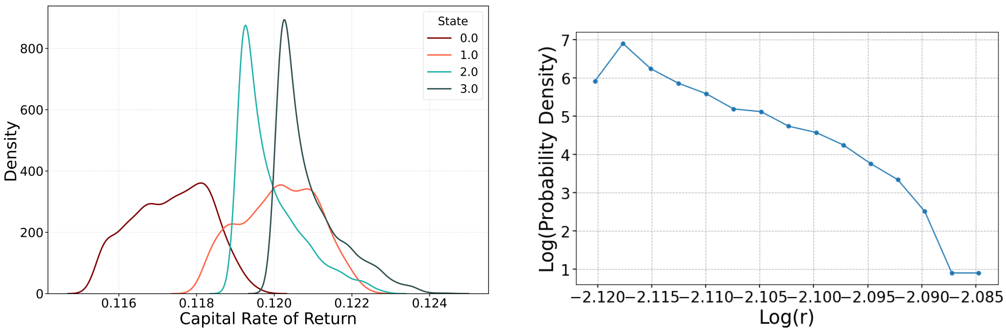
Of note in this diagram is the fact that portfolio volatility both varies by age and, in the context of our model, spikes when agents first enter the market, and when they make the transition from unseasoned workers to seasoned workers between the ages of 30 and 40. This is obviously an artifact of the way we have calibrated unemployment in the model, but the results do suggest that in empirical work on asset markets, demographics matter.

It is useful to compare the portfolio results obtained from the model with those of Brocato and Steed (1998). Brocato-Steed looks at the empirical implications of portfolio optimization based on a conditional CAPM model over the business cycle. They find (see their Table 4) that returns to optimal portfolios based on financial data from 1972 to 1993 are in the 12%-13% range regardless of the business cycle state, but relative standard deviations for these portfolios are about 6% during expansions but 12% during downturns. Our model delivers returns of 12.7% during recessions and 13% during expansions, but the relative standard deviations at only 3.13% and 2.11% respectively. So, while the model matches the returns, and relative volatilities over the phases of the business cycles, it underestimates the volatility numbers.

To show how one can use this modeling approach to study expectations or sentiments issues as they impact the uncalibrated financial and product market outcomes generated by the model, we first list a relevant set of stylized financial market facts (taken from Cont (2001)), where relevant means are not dependent on time-scale considerations. The key facts we consider are

- Absence of autocorrelation
- Heavy tails to return distributions (positive skewness)
- Gain/loss asymmetry
- Leverage effect (inverse correlation of volatility and returns)

In terms of the stylized facts, on the asymmetric gains dimension, Figure 5a shows the empirical distributions of the returns to capital by state. The asymmetry here is striking.



(a) State-Contingent Return to Capital.

(b) Log Tail Distribution.

Figure 5: Capital Distribution by State and Log Tail Distribution in Period 3.

During the recession states, the distribution looks almost uniform, with the increased riskiness of the returns readily apparent. During expansions, the return distribution is positively skewed. If we look at a log-log plot of the tail of the distribution for state 3 (an expansion state) in Figure 10, it looks linear, which is indicative of the tail following a power law.

This is confirmed by a simple regression[9], which yields a slope for the tail of -140, with an R^2 of 0.98. This slope is significantly at odds with empirical estimates of the tail slope between 1 and 3. Finally, on the leverage effect, the model does deliver the inverse relationship between returns and volatility across the business cycle, as noted above. The estimated skewness of the state 3 returns distribution is 1.0, and the estimated kurtosis is 0.4.

In terms of the autocorrelations of returns, Table 3 shows the autocorrelation of the simulated returns to capital at 1, 2, and 3 lags, all of which are quite small.

Autocorrelation of Returns by Lag	
Lag	**Capital**
1	-0.000476
2	-0.000349
3	0.016781

Table 3: Serial correlation coefficients for capital returns at different lags. Values close to zero indicate weak autocorrelation, consistent with market efficiency.

There are a number of other interesting relationships that are apparent in the simulated data. In the interest of brevity, we report on these in online appendix, but Table 4 provides a summary of the data generated by the simulation.

3.3.1. Capital and Labor Shares.

In the simulations, we observe the wage bill $\omega_{h,t}$ and capital holdings $k_{h,t}$ for each age $h = 1, \ldots, 58$, as well as the interest rate r_t for each period t. We calculate output using the per capita production function in (48). Since the model assumes perfect competition, firms earn zero profits in equilibrium.

The capital share associated with age h at time t is

$$\text{capital share}_{h,t} = \frac{r_t k_{h,t}}{\sum_{j=1}^{58} (r_t k_{j,t} + \omega_{j,t})},$$

[9]The regression uses logs of the histogram bin values and frequencies in the tail of the distribution.

Variable	Statistic	State 0	State 1	State 2	State 3	Overall
Output	Mean	20.940	21.055	22.573	22.511	22.000
	SD	1.629	1.596	1.466	1.479	1.693
	Skewness	0.318	0.142	-0.129	-0.074	-0.119
Consumption	Mean	17.529	17.638	17.877	17.906	17.783
	SD	0.140	0.143	0.119	0.126	0.199
	Skewness	-0.013	-0.086	-1.070	-0.875	-0.604
Investment	Mean	3.411	3.417	4.696	4.606	4.217
	SD	1.530	1.491	1.388	1.395	1.552
	Skewness	0.362	0.190	-0.015	0.039	-0.016
Return to Capital	Mean	0.117	0.120	0.119	0.120	0.119
	SD	0.001	0.001	0.001	0.001	0.002
	Skewness	-0.053	0.025	1.370	1.208	-0.532
Bond Price	Mean	0.958	0.962	0.959	0.960	0.960
	SD	0.003	0.002	0.002	0.003	0.003
	Skewness	0.149	0.215	-0.307	-0.270	-0.015
Equity Premia	Mean	7.3%	8.0%	7.7%	7.9%	7.7%
	SD	0.003	0.002	0.002	0.003	0.004
	CV	0.041	0.027	0.025	0. 038	0.052

Table 4: Summary Statistics by State of the 58 Cohort Simulations.

and similarly the labor share associated with age h is

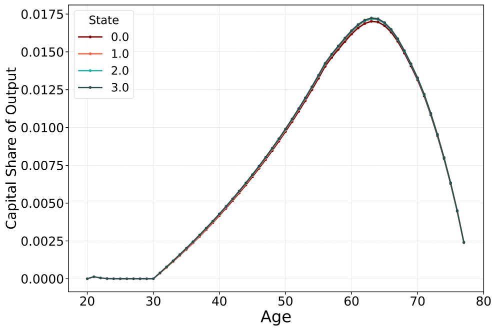
$$\text{labor share}_{h,t} = \frac{\omega_{h,t}}{\sum_{j=1}^{58} (r_t k_{j,t} + \omega_{j,t})}.$$

We then average capital and labor shares by state for each age in Figure 6, panels (a) and (b), respectively. Panel (c) shows the aggregate capital and labor shares:

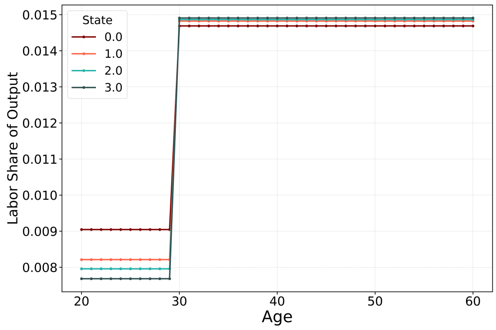
$$\text{capital share}_t = \frac{r_t K_t}{r_t K_t + \sum_{h=1}^{58} \omega_{h,t}},$$

$$\text{labor share}_t = \frac{\sum_{h=1}^{58} \omega_{h,t}}{r_t K_t + \sum_{h=1}^{58} \omega_{h,t}}.$$

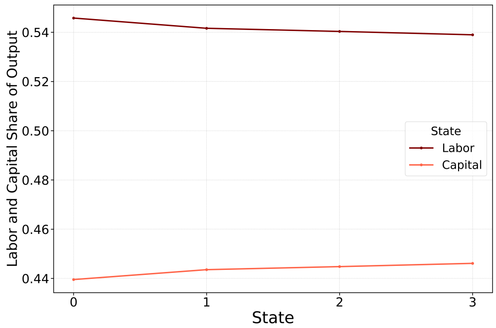
Our results indicate that the capital share of output is procyclical and the labor share of output is countercyclical. Interestingly, when we break down the labor shares by generation, the middle aged generation shares are actually procyclical while the young generation shares remain countercyclical.



(a) Capital Share by Generation.



(b) Labor Share by Generation.



(c) Capital and Labor Share by State.

Figure 6: Capital and Labor Share of Output by State.

3.4. Backing out the expectations function

Given the centrality of expectations in determining the equilibrium stochastic process in the Keynes-Farmer framework, it is worth spending some time asking whether we can characterize the expectations function that gives rise to the Markovian business cycle fluctuations in our model. Farmer et al. (2009) showed how to set up a Markovian expectations function in the context of a standard DSGE model that would then generate demand fluctuations and supply responses consistent with the kind of regime-switching Markovian business cycle fluctuations examined in the empirical literature on these models.

We will show that we get similar results in the SOLG framework based on the characterization of recursive Markovian equilibrium examined by Kim and Spear (2021). This paper shows that when the shocks to the economy are small, a stochastic version of the Grobman-Hartman theorem implies that the equilibrium Markov process is well-approximated by a linear system using the lagged endogenous asset holdings of agents and current shocks as state variables.

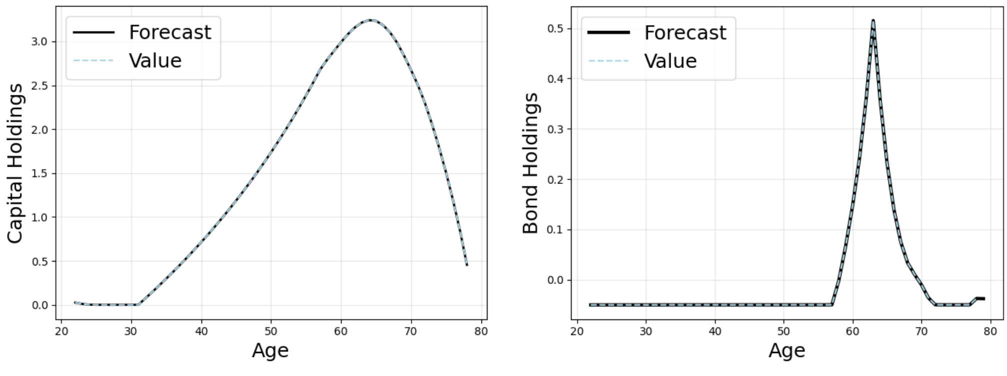
The simulations outlined above give us good reason to assume that the equilibrium process and the expectation function that generate it will be Markovian. Figure 7 shows the power spectra for various endogenous variables, including the simple regime-switching Markov chain that determines the exogenous shocks. Except for differences in actual power, the spectra for each of the variables look quite similar.

To back out the expectation function for the model, we begin by regressing the endogenous state variables – the asset holdings of the agents of each age – on themselves, contingent on the realization of the exogenous shock state. We don't need to worry about explicitly lagging these variables for the regressions since the data are draws from the equilibrium invariant distribution generated by what amounts to a 4-state linear iterated function system (IFS).

The full regressions of all lagged endogenous state variables on their lagged values exhibits serious multicollinearity issues, so we used a ridge regression to isolate the important endogenous states. These turned out to be the capital holdings of agents aged

Figure 7: Power Spectra

11 and 12, together with the bond holdings of agents between the ages of 52 and 56. This result is intuitive. Agents age 11 and 12 are making the transition out of the entry-level labor market into the seasoned labor market and begin active capital accumulation at these ages. Older agents moving out of risky capital into safer bonds constitute the remaining key explanatory variables. Figure 8 shows the average simulated values for capital and bond holdings for all generations compared to the forecast resulting from the ridge regression. For these regressions, the R^2 values are all very close to one, confirming the applicability of the Grobman-Hartman theorem. It is also interesting that the expectations function for the model ends up being fairly parsimonious, and doesn't require agents to forecast from the full 115×115 matrix of lagged asset holdings. The data for this analysis is available in the online supplementary material.



(a) Capital.

(b) Bonds.

Figure 8: Capital and Bond Forecasts from the Estimated Expectations Function.

Beyond the fact that we can back out the expectations function that generates the calibrated outcomes as a rational expectations equilibrium, the coefficient matrices can also provide information relevant to the determination of the equilibrium distributions of returns to capital, bond prices, output or other endogenous variables. As Kim and Spear (2021) note, it is the overlap properties of the equilibrium IFS that determine these distributions. When an IFS satisfies the no-overlap condition (also known in the literature as the open set condition), the support of the state variables will be fractal, and fractal data typically follow power-law distributions. One can see this intuitively by thinking about one side of a Cantor set, shown in Figure 9.

Figure 9: Cantor Set

In the Cantor set, the points with positive mass accumulate against the endpoints of the unit interval, while the self-similarity of the Cantor set generates the power-law tails of the distribution.

Verifying the no-overlap condition in high-dimensional systems is quite difficult, but we can gain some insight into the possibility that the condition holds by computing the state-contingent spectral radius for each of the coefficient matrices. The spectral radius is defined as the absolute value of the largest eigenvalue (in absolute value terms) of the matrix. A result of Falconer (1988) shows that the spectral radius of the largest operator in a contractive linear iterated function system gives an upper bound on the Hausdorff dimension of the equilibrium invariant set for the model using the formula

$$H \leq -\frac{ln\, p}{ln\, \alpha^*}$$

where α^* is the largest spectral radius of the p-component linear IFS. This is derived in the online appendix. For the two-state phase-contingent iterated function systems generated by the model, we have $\alpha^*_{expansion} = 0.18$ and $\alpha^*_{recession} = 0.17$. These yield bounds on the Hausdorff dimension of $H \leq 0.4235$ for expansions and $H \leq 0.3912$ for recessions.

These results indicate that the distribution of asset holdings is fractal. As we noted above, the fractal dimension for expansions can explain the observed power function tail distribution of the returns to capital or of output, while an absolutely continuous distribution of returns – which is not ruled out by the fractal nature of the iterated function system – during recessions would explain the more uniform distributions observed for these states. This calculation also indicates that the match the observed power law tails in the data, we would need the spectral radius during expansions to be closer to $\frac{1}{3}$.

3.4.1. Comments

Studying the rational expectations functions generated by the equilibrium of the model is important for several reasons.

First, the fact that expectations in SOLG environments must include lagged endogenous state variables, while these are not required in conventional DSGE approaches provides a simple empirical test of model specification for researchers studying economic expectation formation. A related issue arises from the fact that forecasts of the type generated by the model are a concrete manifestation of more general consumer or producer sentiment. Psycho-economic theories of sentiment formation necessarily require specification of how emotion translates into forecasts, so some understanding of the nature of forecasts in dynamic economic activities is useful.

A second important issue arises from the fact of the multicollinearity in the full set of regressors for the forecast mappings. While the econometric technique of Ridge regression reduces the multicollinearity problem, it also identifies the key economic considerations that determine the REE forecasts: the transition point where young unseasoned workers become seasoned, and the transition point where older households shift from equity to bonds for their savings.

The linearity of the expectations function also suggests a mechanism for examining what happens during transitions between one economic regime and another. Different regimes can arise because of policy changes, or when expectations stabilize in a particular configuration, such as a depressed steady state. We could model this, for example, by calibrating the production side of the economy to model the features of the Great Depression. To the extent that the aggregate shocks in this version of the model are small (so that being stuck in a depressed state is a deterministic function of demand deficiency), the stochastic Grobman-Hartman theorem will apply and the expectations functions will be linear. One can then simulate the transition from the normal business cycle regime to the depressed regime by defining a convex linear homotopy from one regime to the other and then running simulations based on these expectations. While the interim expectations will likely not be rational expectations, there is no compelling reason to believe these transitional states should satisfy the REE fixed point criterion. Finally, this procedure is an alternative to the widely-used process of examining impulse response functions, and it has the benefit of actually producing simulated time-series that can be compared with actual data.

We note also that the Kim-Spear results extend the basic applicability of the Grobman-Hartman theorem to general SOLG models with heterogeneous agents. In particular, the fact that the aggregated expectations matrices have their eigenvalues given by the stable eigenvalues for the full equilibrium dynamic system (as in Kehoe and Levine (1984)), means that the analysis above of the possibility of fractal supports for the conditional equilibrium probability measures, and the ability to simulate non-REE transition dynamics remains possible.

3.4.2. Economic Interpretation

The economic question posed by this data is why the symmetric Markov process of employment shocks gives rise to a skewed distribution of returns during expansions, but a more uniform distribution during downturns. As we noted above, because the recursive updating of the wealth distribution during expansions is fractal, the high-dimensional clouds of wealth combinations will have outliers that occur with relatively large probability, relative to a corresponding normal density. We believe this is related to the observed negative skewness of the conditional output distribution. Figure 10 shows the two state-contingent distributions of output for states 0 (recession) and 3 (expansion). The state 3 distribution has a skewness coefficient of -0.15 and is clearly leptokurtotic. The concentration of outputs at the high end will yield a concentration of relative lower returns (given that these are just the marginal products of capital), while the negatively skewed lower tail will generate a positively skewed upper tail to the distribution of returns.

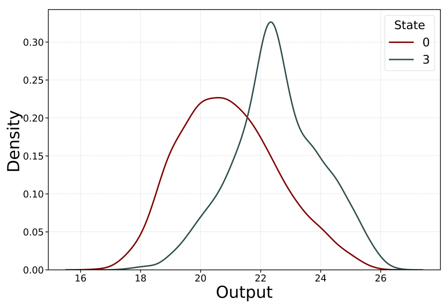
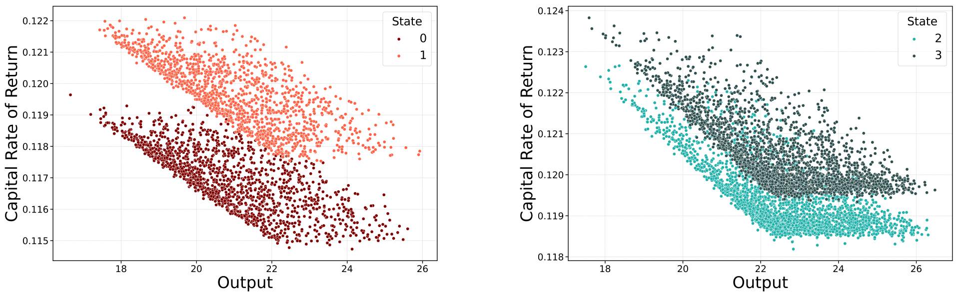


Figure 10: State 0 vs. State 3 Output Distribution

The reason for the negative skewness of the output distribution during expansions can be explained by observing that savings do not vary much over the course of the business cycle (see Figure 2 above), so that young agents born into recession states, or getting repeated recession states while young, will save rather than consume, leading to instances of consumption demand deficiency during upturns. Finally, Figure 11 shows the relationship between the return to capital at time $t-1$ and output at time t (which includes undepreciated time $t-1$ capital) during downturns and upturns.



(a) Output-Return Relation: Recessions.

(b) Output-Return Relation: Upturns

Figure 11: Output-Return Relation During Recessions and Upturns.

Both scatter plots show a lower envelope where output and returns are negatively related. This reflects the fact that the marginal product of capital is decreasing in output when labor inputs are fixed. The relationship manifests itself at the lower envelope since this corresponds to situations when capital is minimized for the resulting output realization. There is also clearly a level of output beyond which the return to capital is unchanged. This reflects a situation in which output increases due to correlated changes in both employment and capital that keep the return to capital fixed. We note also that the clustering of realizations of output and returns to capital near the lower envelope is more pronounced during expansions than during recessions. We conjecture that this is due to more efficient use of capital during upturns relative to downturns.

4. Conclusion

This paper studies Keynesian search in a stochastic overlapping-generations economy. The central result is that Farmer-style labor-market incompleteness continues to generate steady-state indeterminacy, but the belief functions that support equilibrium are disciplined by the endogenous evolution of the wealth distribution. In a representative-agent environment, a belief rule can often be specified directly, and any candidate rule constitutes a rational-expectations equilibrium by construction. In an SOLG environment, this arbitrariness is no longer available: rational expectations require the forecast rule, temporary equilibrium allocations, and the induced wealth distribution to form a fixed point, so an admissible belief function must reproduce the wealth distribution it presupposes. This consistency requirement restricts which belief functions are admissible though it does not by itself resolve the Farmer-style steady-state indeterminacy. In our calibrated model, the steady state is closed by our assumptions on the exogenous unseasoned-employment profile across regimes and by setting the unseasoned wage equal to the marginal product of unseasoned labor. The fixed-point restriction then determines which expectations are consistent with the resulting equilibrium, rather than determining the equilibrium allocation itself.

The quantitative 58-period model shows how this mechanism can be implemented in an annual-frequency lifecycle economy with entry-level search, matching-efficiency shocks, and endogenous portfolio dynamics. The neural-network solver provides a practical method for computing the recursive rational-expectations fixed point in this high-dimensional environment. The simulations also illustrate how matching-efficiency shocks can generate lifecycle, asset-market, and business-cycle dynamics without relying exclusively on aggregate TFP shocks.

Several extensions remain. We have noted at a number of different places that the results we obtain do not fully match the data. Future work on getting the modeling framework to better match the data would likely entail developing a more sophisticated model – one with more flexible production functions, more flexible (Epstein-Zin) preferences, endogenous labor supply choices, a fully dual labor market across the whole working population and relaxing the competitive markets assumption to allow for more flexible market structures (e.g.. working with Shapley-Shubik or Dixit-Stiglitz models) – to really check whether a model of this type can match the data. Part of our purpose in developing this analysis to begin with is to lay out a foundation on which future research can build.

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